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Chapter 3

GELFAND-P000208

[source number: 1]

1

Year 1967. Gelfand Seminar.

18/IX. Kirillov.

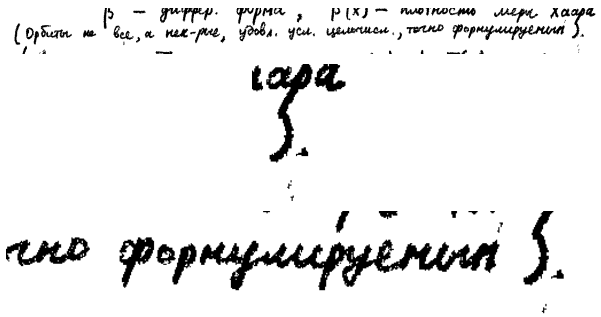
Formula (without proof or counterexamples) for characters of irreduc. repres. of Lie groups.

$$\chi(x) = p(x)^{-1/2} \int_{\Omega} e^{i\langle F,x\rangle}\beta \qquad (\beta = \beta(F))$$

$x \in \mathfrak{g}$, $F \in \mathfrak{g}^*$, Ω — an orbit in $\mathfrak{g}^*$ under G,

β — a differ. form, $p(x)$ — the density of Haar measure.

(Not all orbits, but certain ones satisfying integrality cond., precisely [formulated: ending -емый / -емым?]}.



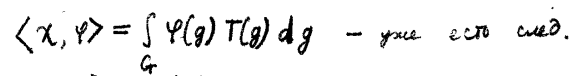
Uncertain reading. The ending appears to be -емый. The final stroke also permits -емым. A curly brace and a point follow.

Original source details: GELFAND-P000208

(G — a group, T — a unitary repres., $\chi(g) = \operatorname{tr} T(g)$ (in finite dim.)

In the general case χ — a gen. function

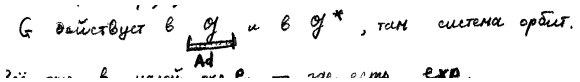
$$\langle \chi, \varphi \rangle = \int_G \varphi(g) T(g)\, dg \quad \text{— already has a trace.}$$

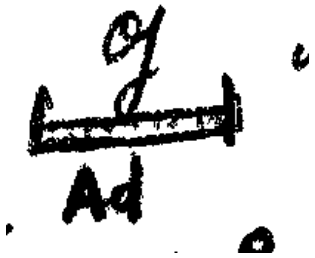


Original source details: GELFAND-P000208

Manuscript notation. The equation has no trace sign before the integral. The text after the integral says “already has a trace.”

(Such a def. of the char.).)





Original source details: GELFAND-P000208

Diagram and typed description.

G acts in $\underset{\mathrm{Ad}}{\mathfrak{g}}$ and in $\mathfrak{g}^*$; there is a system of orbits there.

All this is in a small nbhd of e — where there is exp.

β is constructed from the struct. const. of the algebra.

[Crossed out: illegible.]



Original source details: GELFAND-P000208

Undeciphered passage. One word beneath dense repeated loops. Undetermined candidates: “Предложения” (Proposals) / “Продолжение” (Continuation). The letters beneath the strokes are illegible.

Planned.

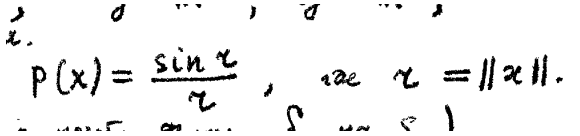
1. Expository talks on logic. (including the solution of the continuum hypothesis)
2. Lidskii “Asymptotics of eigenv. of self-adj. oper.”
3. Work of Kirillov; Kazhdan; Graev.

Example. $G = SO(3)$, $\mathfrak{g} = \mathbb{R}^3$, $\mathfrak{g}^* = \mathbb{R}^3$,

the action — rotations.

Orbit — S^2.

$$p(x) = \frac{\sin \tau}{\tau}, \qquad \text{where } \tau = \|x\|.$$

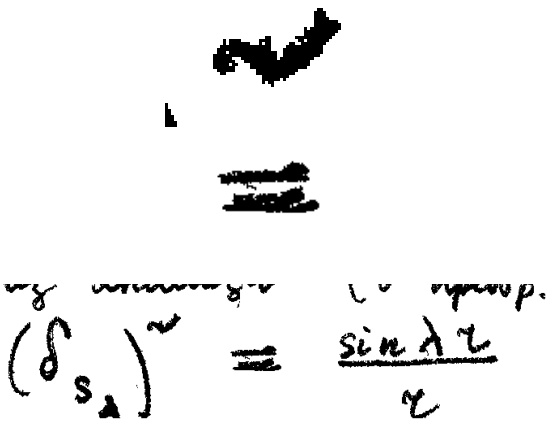


Original source details: GELFAND-P000208

Uncertain reading. The three occurrences have a hooked top and a bent stem. The provisional reading is τ. An alternative is r. The fraction has no visible power.

A fact from analysis (on the Fourier transform of δ on S_λ)

$$(\delta_{S_\lambda})^{\sim} \equiv \frac{\sin \lambda\tau}{\tau}$$



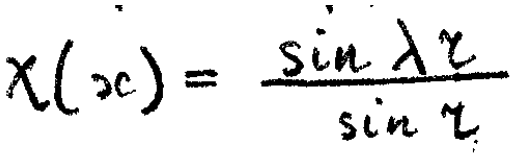
Uncertain reading. A wavy stroke $\sim$ sits above the parenthesis. The lower strokes of the equality sign run together. The provisional reading is $\equiv$. An alternative is $=$. The radial variable is the same.

Original source details: GELFAND-P000208

GELFAND-P000209

β — ordinary surface [measure] on the sphere.
Answer.

$$\chi(x) = \frac{\sin \lambda\tau}{\sin \tau}.$$



Uncertain reading. The radial grapheme is the same as on the preceding page: τ / r. Candidate in the ratio: τ; the literal letter is undetermined.

Original source details: GELFAND-P000209

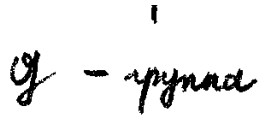
All this for $\lambda = 2n + 1$ (countably many repres.)
$\beta(x)$ — $\int$ over cycles of the orbit of certain integral forms.
Work of Fuchs and Gelfand.
Fuchs. Study of principal bundles
[Crossed out: illegible.]



Undeciphered passage. A short, densely crossed-out entry between “bundles” and “with fiber”. The letters are undetermined.

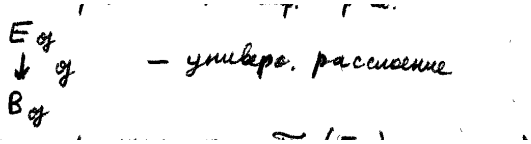
Original source details: GELFAND-P000209

with fiber G (top. group)
Types of invar. — as top. spaces and classif. spaces.
$\mathfrak{g}$ — a group



Manuscript notation. The loop and descender match $\mathfrak{g}$ on the preceding page. The earlier phrase “with fibre” uses a capital G.

Original source details: GELFAND-P000209



Original source details: GELFAND-P000209

Diagram and typed description.

$E_{\mathfrak{g}}$
$\downarrow \mathfrak{g}$ — universal bundle
$B_{\mathfrak{g}}$

($E_{\mathfrak{g}}$ [Crossed out: illegible.]



Original source details: GELFAND-P000209

Undeciphered passage. One word beneath repeated dense strokes before “aspherical”. The letters are undetermined.

aspherical, i.e. $\pi_k(E_{\mathfrak{g}}) = 0$, $k > 0$).

A char. class is determ. by specifying $\alpha \in H^q(B_{\mathfrak{g}}; A)$

(Maps of principal bundles [commute?] with the group action).



Original source details: GELFAND-P000209

Uncertain reading. The angular strokes at the end of the abbreviation are joined. The Russian reading remains перестан. or перестав.

Analogously, instead of H^q — any contravar. functor h to the category of sets.

All char. classes — $h(B_{\mathfrak{g}})$

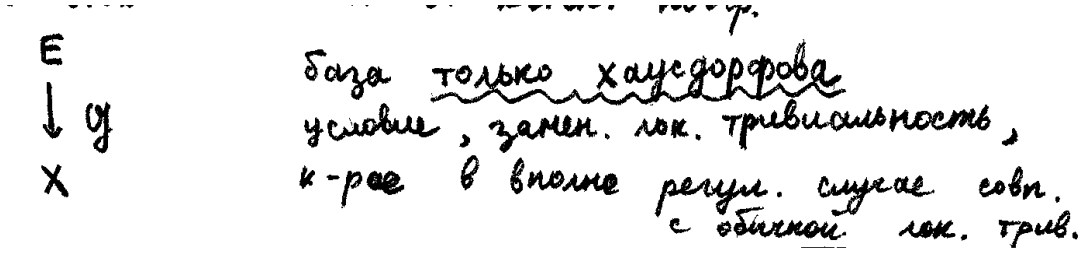
(this is if h — a homotopy functor)

$\hat{\mathfrak{g}}$ — a max. comp. subgroup

(all of them are homot. equiv. and their classif. spaces hom. equivalent.)

and equiv. to $\mathfrak{g}$.

G. Problem: find the actual classif. space of noncomp. groups, without identifying it with the max. comp. subgr.



Original source details: GELFAND-P000209

Typeset diagram

E
$\downarrow \mathfrak{g}$
X

base only Hausdorff
a condition replacing loc. triviality,
which in the completely regul. case coincides
with usual loc. triv.

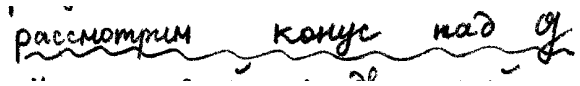
GELFAND-P000210

[source number: 2]

2

The following property is required to hold:

consider the cone over $\mathfrak{g}$



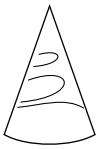
Original source details: GELFAND-P000210

Manuscript notation. The group glyph has the same loop and descender as $\mathfrak{g}$ on the preceding page.



Original source details: GELFAND-P000210

Uncertain reading. The outer cone outline, rounded base, and internal curved paths. Closure of the paths and possible arrow directions are undetermined.



$\mathfrak{g}$ acts on it with one fixed point.
$\forall x \in X$, $\exists$ a map $E \longrightarrow C\mathfrak{g}$, commuting with the group action;



Original source details: GELFAND-P000210

Manuscript notation. The manuscript repeats перестан. before “with the group action.”

the fiber over [Crossed out: an illegible sign.]



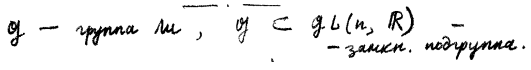
Original source details: GELFAND-P000210

Undeciphered passage. Before x a sign with a lower loop is densely crossed out. Undetermined candidates: y / g.

x maps isomorphically onto the base of the cone
(Local triviality follows from this).
(For a completely regul. base (there exists a function 1 at the point, 0 outside a given nbhd) loc. triviality $\Longrightarrow$ this property.)
(but in this case the theory becomes trivial and all computed invar. are trivial)

D. Kazhdan’s remark: then the height of the cone determines a cont. function on X.
So this def. is incorrect.

$\mathfrak{g}$ — a Lie group, $\mathfrak{g} \subset gL(n, \mathbb{R})$ — a closed subgr.

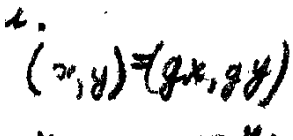


Original source details: GELFAND-P000210

Manuscript notation. The first letter before L has a loop and descender. The manuscript writes $gL(n, \mathbb{R})$.

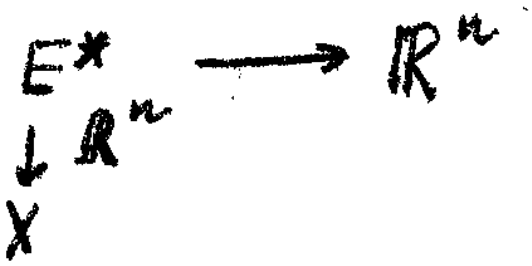
(i.e. $\mathfrak{g}$ acts on $\mathbb{R}^n$).
There exists a construction which makes a vector bundle (induced) from a principal bundle

$E^* = E \times \mathbb{R}^n/\mathfrak{g}$.
$(x, y) = (gx, gy)$



Original source details: GELFAND-P000210

Manuscript notation. Two strokes form $=$ before the second pair and touch its opening parenthesis.



Original source details: GELFAND-P000210

Diagram and typed description.

$$\begin{array}{ccc} E^* & \longrightarrow & \mathbb{R}^n \\ \downarrow & & \\ X & & \end{array}$$

There must exist a map $E^* \longrightarrow \mathbb{R}^n$,
[Crossed out: illegible.]



Original source details: GELFAND-P000210

Undeciphered passage. A short word before “linear” is crossed out by two strokes. The letters are undetermined.

linear on the fibers and
such that at a certain point x_0 (given)
it — an isomorphism.
(No compatibility with the group action).



this is the def. of loc. triviality (for Hausd.) and this is precisely the requirement.
Kazhdan’s remark cannot be repeated.

GELFAND-P000211

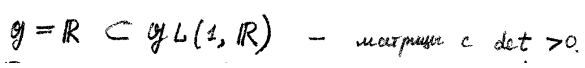
E is recovered from E^*.
One cannot construct a classif. space, but there is an analogue.
Example:
[An undetermined thickened sign.]



Original source details: GELFAND-P000211

Uncertain reading. Before the group notation is a separate thickened sign with a lower loop and a transverse stroke: “в” (in) / a crossed-out earlier sign. Neither candidate is established.

$\mathfrak{g} = \mathbb{R} \subset gL(1, \mathbb{R})$ — matrices with det > 0.



Original source details: GELFAND-P000211

Manuscript notation. The same looped letter with a descender as on manuscript page 2 precedes L. The manuscript writes gL.

A nontrivial $\mathbb{R}$-bundle (also "universal").
(principal)

The direct product of countably many lines in the Tych. topology.

$$(x_1, x_2, \ldots) \quad \text{— sequences}$$

$$\alpha(x_1, \ldots) = (x_1 e^{\alpha}, \ldots)$$

The quotient space S cannot be embedded in E (!) — — there are no sections.

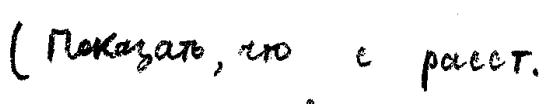
Fact: in $\mathbb{R}^{\infty}$ one cannot choose one point on each ray.

Proof.

S is Hausdorff, but every cont. function on it is const,

so such a map — to a point (coords. — cont. functions).

(Show that with the distance

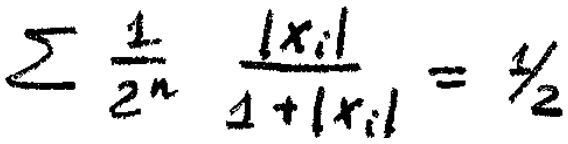


Original source details: GELFAND-P000211

Manuscript notation. The manuscript writes расст., an abbreviation for "distance." The two c letters are followed by т.

$$\sum \frac{1}{2^n} \frac{|x_i|}{1 + |x_i|} = 1/2$$

Manuscript notation. The weight is written with 2^n. Both occurrences of x have subscript i.



Original source details: GELFAND-P000211

one obtains a discontinuous map.)

Suppose there is f. (a cont. function).

$$(x_1, x_2, \ldots)$$

$$a_i = f(\underbrace{0, \ldots, 0}_{i}, 1, \ldots, 1, \ldots)$$

$$a = f(x_1, x_2, \ldots)$$

We shall prove that $a = \lim_{i \to \infty} a_i$.

$\forall \varepsilon > 0$ $\exists$ a nbhd of the point

[Crossed out: an illegible sign.]



Original source details: GELFAND-P000211

Undeciphered passage. Before the coordinate parenthesis a sign is densely crossed out. The original grapheme is undetermined.

$(x_1, x_2, \ldots) \in S$,
where the difference is $< \varepsilon$. (from a).

$$(x_1^0, x_2^0, \ldots)$$

(A neighborhood — if the preimage is open)

GELFAND-P000212

[source number: 3]

3

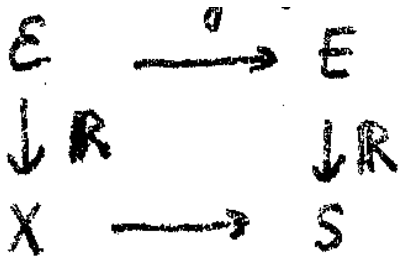
A neighborhood in E

$$|x_1 - x_1^0| < \delta$$
$$\ldots$$
$$|x_N - x_N^0| < \delta.$$

Keep these coordinates the same and make the others very large.

$$\left(\frac{x_1^0}{M}, \ldots, \frac{x_N^0}{M}, 1, 1, \ldots\right),$$

and these approximate the point $(0, \ldots, 0, 1, \ldots, 1, \ldots)$.
A map always exists (but is not unique)



Original source details: GELFAND-P000212

Uncertain reading. The upper-left vertex is rounded: calligraphic $\mathcal{E}$ / ε. Provisional reading: $\mathcal{E}$; the letter is undetermined. Both horizontal arrows point right, both vertical arrows down with labels $\mathbb{R}$.

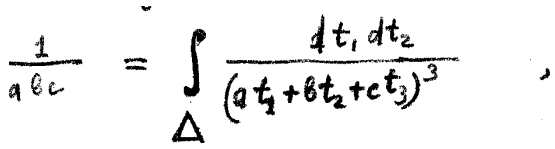
$$\begin{array}{ccc} \mathcal{E} & \longrightarrow & E \\ \downarrow{\scriptstyle \mathbb{R}} & & \downarrow{\scriptstyle \mathbb{R}} \\ X & \longrightarrow & S \end{array}$$

S is constructed from a product of $\mathbb{R}$ of any cardinality (depends on X)
$25/\overline{\text{IX}}$.
A good formula (Feynman)

$$\frac{1}{ab} = \int_0^1 \frac{dt}{(at + b(1-t))^2}$$

(decomp. into partial fractions)

$$\frac{1}{abc} = \int_{\Delta} \frac{dt_1\, dt_2}{(at_1 + bt_2 + ct_3)^3},$$



Manuscript notation. The integral immediately follows the equals sign. The manuscript shows no numerical multiplier.

Original source details: GELFAND-P000212

Δ — a simplex

$$t_1 + t_2 + t_3 = 1, \quad t_i \geq 0.$$

Next time Fuchs and then logic.

There is a counterexample to the hypothesis of a common fixed point of commuting cont. maps of an interval, and it is proved that if they are differentiable, there is a point fixed for one and of finite multiplicity for the other.

GELFAND-P000213

Fuchs (continuation).

$\mathfrak{g} \subset gL(n;\mathbb{R})$



Manuscript notation. Both group letters have a loop and a descender. A semicolon follows n.

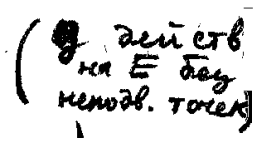
Original source details: GELFAND-P000213

principal bundles

$$(\xi) \qquad p : E \longrightarrow X$$
$$(\xi^{\#}) \qquad p^{\#} : E^{\#} \longrightarrow X$$

(X — Hausdorff)

($\mathfrak{g}$ acts on E without fixed points]



Uncertain reading. The right-hand note ends with an upright stroke that hooks inward. It is provisionally set as].

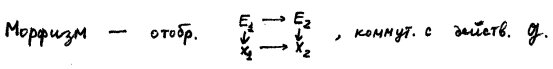
Original source details: GELFAND-P000213

(

$$xg_1 = xg_2 \Rightarrow$$
$$\Rightarrow g_1 = g_2$$

Loc. triviality: $\forall x \; \exists$ a map $E^{\#} \longrightarrow \mathbb{R}^n$, an isom. at the point x, linear on the fibers.

(One cannot require 0 outside small nbhds, since then even $X \times \mathbb{R}^n$ may fail to qualify because of nonregularity)



Original source details: GELFAND-P000213

Diagram and typed description.

A morphism — a map

$$\begin{array}{ccc} E_1 & \longrightarrow & E_2 \\ \downarrow & & \downarrow \\ X_1 & \longrightarrow & X_2 \end{array}$$

, commuting with the action of $\mathfrak{g}$.

Examples of functors:

1) cohomology

2) H-bundles (H — another group)

correspondingly charact. classes.

$h_{\text{alg}}(\mathfrak{g})$ — the set (group) of all char. classes constructed from the functor h.

1) For any top. space

[Crossed out: an illegible sign.]



Original source details: GELFAND-P000213

Undeciphered passage. After “пр-ва” (space), before X, an additional sign is densely crossed out. The original grapheme is undetermined.

X, the sheaf $\mathbb{R}$ of germs of cont. real functions on X is defined.

Put

$$h(X) = H^q(X, \mathbb{R}).$$

(for [Crossed out: illegible.]



Original source details: GELFAND-P000213

Undeciphered passage. A fragment between “for” and “paracomp.” is crossed out; a descender is visible. The letters are undetermined.

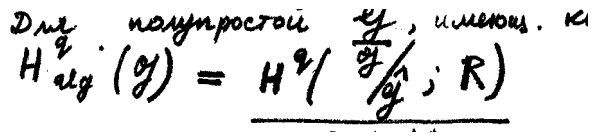
paracomp. X this is 0)

There is defined

$$H^q_{\text{alg}}(\mathfrak{g})$$

Answer. For semisimple $\mathfrak{g}$ having finitely many connected components

$$H^q_{\text{alg}}(\mathfrak{g}) = H^q(\bar{\mathfrak{g}}/\hat{\mathfrak{g}}; R)$$



Original source details: GELFAND-P000213

Manuscript notation. A straight bar marks the first group glyph and a hat marks the second. The descender on the line above touches the bar.

with coeffs. in a constant sheaf —
— real coeffs.

GELFAND-P000214

[source number: 4]

4

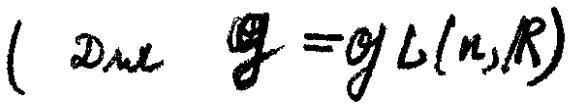
Here $\bar{\mathfrak{g}}$ — a compact form of $\mathfrak{g}$
$\hat{\mathfrak{g}}$ — a max. comp. subgroup
[Crossed out: $\bar{\mathfrak{g}}$?]



Original source details: GELFAND-P000214

Undeciphered passage. Before the final group sign an entry is densely crossed out. Candidate $\bar{\mathfrak{g}}$ from the visible upper bar and lower loop; the precise form beneath the strokes is undetermined.

$\mathfrak{g}$. (For $\mathfrak{g} = gL(n, R)$



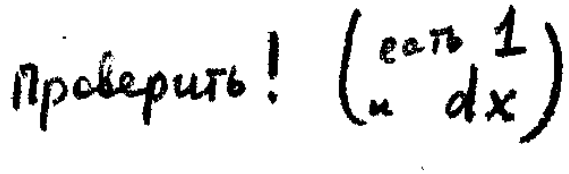
Original source details: GELFAND-P000214

Manuscript notation. The letter before L repeats the loop and descender on the preceding page. The manuscript writes gL.

there will be $\bar{\mathfrak{g}} = U(n)$, $\hat{\mathfrak{g}} = O(n)$
$\bar{\mathfrak{g}}$ is obtained as a max. comp. subgroup of the complexification of $\mathfrak{g}$)
2nd order: as for S^2.
For a compact group everything is trivial.
For a nonsemisimple one, take $\mathfrak{g}/\hat{\mathfrak{g}}$ and differ. forms on $\mathfrak{g}/\hat{\mathfrak{g}}$ invariant under $\mathfrak{g}$.
d(invar.) — invar. And the homologies of this complex are isomorphic to $H^q_{\text{alg}}(\mathfrak{g})$.

$$H^q_{\text{alg}}(\mathbb{R};\mathbb{R}) = \begin{cases} R & q = 0,1 \\ 0 & q > 1. \end{cases}$$

Check! (there are 1 and dx)



Manuscript notation. The "Check!" note sits to the right of the formula, with 1 above dx inside parentheses.

Original source details: GELFAND-P000214

2) X, H — a top. group

$EH(X)$ — the set of all loc. triv. (in the usual sense) bundles (principal) on X.

\

Theorem.

$$EH_{\text{alg}}(\mathfrak{g}) = \text{Hom}(\mathfrak{g}, H)$$

(cont. hom. [Crossed out: an illegible sign.]



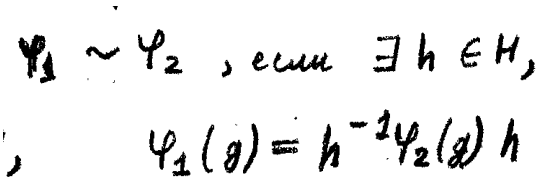
Undeciphered passage. After "cont. hom." are doubled curved and oblique strokes: a cancelled) / the sign φ. The reading is undetermined.

Original source details: GELFAND-P000214

, more precisely, classes of conjugate homom.)

The natural answer — a hom. $\mathfrak{g} \longrightarrow H$ corresponds to a char. class, naturally defined.

$$\varphi_1 \sim \varphi_2, \text{ if } \exists h \in H, \qquad \varphi_1(g) = h^{-1}\varphi_2(g)h$$

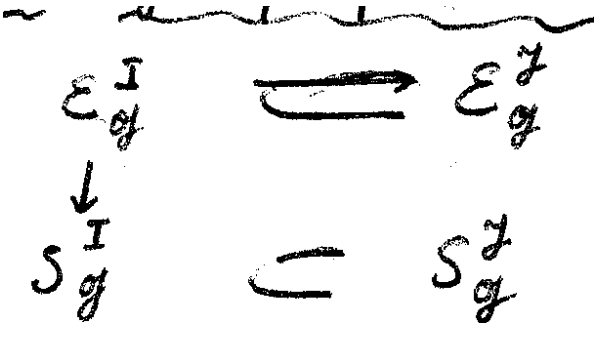


Uncertain reading. The map glyphs in the formula have a closed right loop and an open left arm. The reading φ is preferred.

Original source details: GELFAND-P000214

GELFAND-P000215

Construction of a universal bundle.



Uncertain reading. The upper stroke ends in a rightward arrowhead. The lower stroke curves upward on the left. The redraw follows both strokes as drawn. The rounded upper letters are rendered as $\mathcal{E}$.

$$\begin{array}{ccc} \mathcal{E}^I_{\mathfrak{g}} & \rightleftharpoons & \mathcal{E}^J_{\mathfrak{g}} \\ \downarrow & & \\ S^I_{\mathfrak{g}} & \subset & S^J_{\mathfrak{g}} \end{array}$$

Original source details: GELFAND-P000215

$I \subset J$ — arbitrary sets.

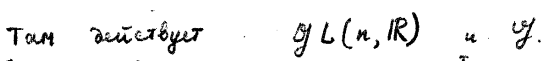
Any bundle maps into a universal one with I of sufficiently large cardinality.

Consider $\mathbb{R}^I \ni (x_i,\ i \in I)$.

The topology — Tychonoff.

and $\underset{n \text{ lin. indep. vect.}}{n\text{-frames}}$ in it — this is precisely $\mathcal{E}^I_{\mathfrak{g}}$.

$gL(n,\mathbb{R})$ and $\mathfrak{g}$ act there.



Original source details: GELFAND-P000215

Manuscript notation. The looped letter with a descender recurs before L. It is set as gL.

The quotient space is precisely our $S^I_{\mathfrak{g}}$.

Observe that $\mathcal{E}^I_{\mathfrak{g}}$ does not depend on $\mathfrak{g}$.

Lemma. $\forall \mathcal{E}^I_{\mathfrak{g}}$ and a cont. function f on it $\exists$

$J \subset I$ such that $I \setminus J$ is at most countable, and

and $f|_{\mathcal{E}^J_{\mathfrak{g}}} = \text{const}$.

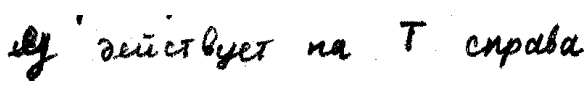
GELFAND-P000216

[source number: 6]

$2/\overline{\text{X}}$. Fuchs.

T — a space, $S \subset \text{Map}(T,T)$, compact open top.

$\mathfrak{g}$ acts on T on the right



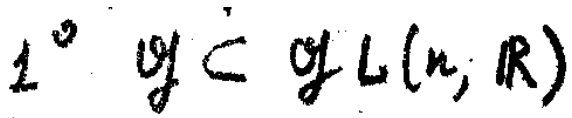
Original source details: GELFAND-P000216

Manuscript notation. The group sign has a loop and descender. Here it differs from the simpler g in the next equation.

$$f \in S, \quad g \in \mathfrak{g} \Rightarrow fg \in S,\ gf \in S.$$

Examples.

1°. $\mathfrak{g} \subset gL(n;\mathbb{R})$,



Original source details: GELFAND-P000216

Manuscript notation. The letter before L has a loop and descender. A semicolon follows n. The left stroke of $\mathbb{R}$ is doubled.

$T = \mathbb{R}^n$; S — all lin. maps (including zero maps).

2°. $\mathfrak{g}$ — a Hausdorff top. group, $T = C\mathfrak{g}$ — the cone over $\mathfrak{g}$

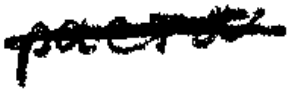


Original source details: GELFAND-P000216

Diagram and typed description.



$\mathfrak{g}$ acts naturally, S is generated by them and similarities [Crossed out: растя... / растяже...? (dila...?)]



Undeciphered passage. Before “contractions” a word or beginning of a word is crossed out; the opening “раст...” is distinguishable. The ending is undetermined.

Original source details: GELFAND-P000216

contractions toward the vertex (and to a point).

Axioms.

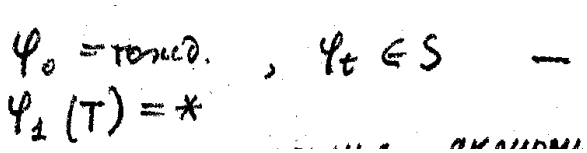
1°. T is contractible by transformations in S to a marked point $*$.

2°. $\mathfrak{g}$ is closed in the group of all homeom. of T to itself.

3°. $*$ is fixed under all transf. in S.

4°. $\varphi_t : T \longrightarrow T$

$$\varphi_0 = \text{identity}, \quad \varphi_t \in S \quad —$$
$$\varphi_1(T) = *$$



Manuscript notation. The map sign has an open left arm and a closed right loop. It reads φ.

Original source details: GELFAND-P000216

— the homotopy of axiom 1°.

Then there exist φ_t such that φ_t for $t < 1$ are topological embeddings.

5°. S contains all transf. of $\mathfrak{g}$, and maps close to maps of $\mathfrak{g}$ in S are top. embeddings.

$U(1) \subset S$ consists of embeddings.

In this situation spaces are defined

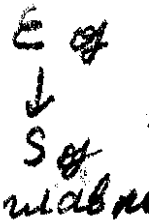


Diagram and typed description.

$$\begin{array}{c} \mathcal{E}_{\mathfrak{g}} \\ \downarrow \\ S_{\mathfrak{g}} \end{array}$$

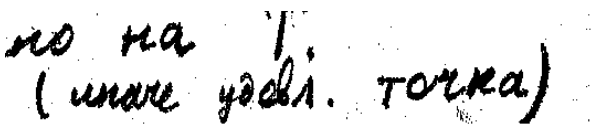
, the action of $\mathfrak{g}$ on $\mathcal{E}_{\mathfrak{g}}$ — a universal principal bundle.

Original source details: GELFAND-P000216

$\underbrace{T \times \cdots \times T}_{I \text{ times}}$ Tych. topology.

6°. $\mathfrak{g}$ acts effectively on T,

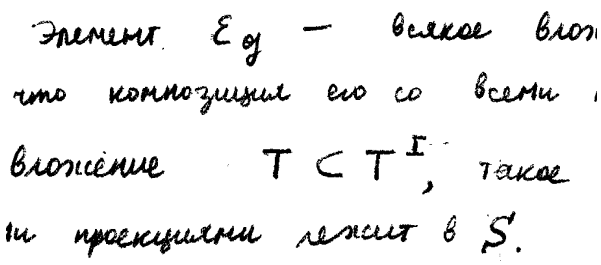
(otherwise [удвл.?] point)



Uncertain reading. The parenthesis provisionally reads “удвл.”. The expansion is undetermined.

Original source details: GELFAND-P000216

An element of $\mathcal{E}_{\mathfrak{g}}$ — any embedding $T \subset T^I$ such that its compositions with all projections lie in S.



Manuscript notation. The subscript of the rounded $\mathcal{E}$ has the same loop and descender as the group sign on the preceding page.

Original source details: GELFAND-P000217

1) $T = \mathbb{R}^n$ — all n-frames.

2) $T = C\mathfrak{g}$, $\mathcal{E}_{\mathfrak{g}} = C\mathfrak{g} \times \ldots \times C\mathfrak{g} \setminus \{*, \ldots, *\}$.

$\mathfrak{g}$ acts on $\mathcal{E}_{\mathfrak{g}}$ by multiplication by transf. in $\mathfrak{g}$.

I finite $\Rightarrow$ 1) the usual classif. space
2) Milnor's constr. with a not very substantial modification.



Diagram and typed description.

$\mathfrak{g} = \mathbb{R}$, $T =$ •————

Original source details: GELFAND-P000217

$\mathcal{E}_{\mathfrak{g}}$ — sequences of nonneg. numbers.

For each T its own definition of loc. triviality.

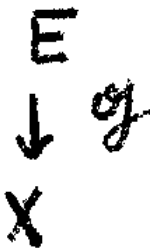


Diagram and typed description.

$$\begin{array}{c} E \\ \downarrow \mathfrak{g} \\ X \end{array}$$

Original source details: GELFAND-P000217

is loc. triv. if the induced bundle



Diagram and typed description.

$$\begin{array}{c} E^{\#} \\ \downarrow T \\ X \end{array}$$

Original source details: GELFAND-P000217

, has the property: $\forall x \; \exists$ a map $E^{\#} \longrightarrow T$, a homeom. on the fiber over x, and on $\forall$ fiber it is a transf. in S.

Theorem.

Every bundle maps into this one.

(X — all Hausdorff)

Defined $H^q_{\text{alg}}(\mathfrak{g};\mathbb{R})$

Theorem.

Whatever T and S may be, $H^q_{\text{alg}}(\mathfrak{g},\mathbb{R})$ does not depend on their choice

(Instead of $\mathbb{R}$ one may take a group A)

What will there be?

Uspenskii, Lectures on logic 2

Lidskii, Asymptotics of eigenfns. (after Hörmander)

Sinai, Conditions on diffeom. under which there is an invar. measure

Kazhdan, Discr. groups — solution of Selberg's problem

Kushnirenko, Analyt. action of a semisimple group in a nbhd. of a fixed point — linear

GELFAND-P000218

[source number: 7]

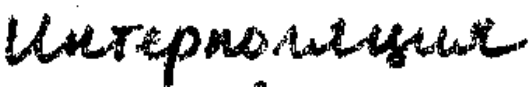
Tyurina and S. Gelfand survey of Pham's works.

Anosov and Maslov

I. Bernstein P^λ

Gelfand, Graev

S. G. Krein Interpolation



Uncertain reading. The final strokes favour the singular "Интерполяция" (Interpolation).

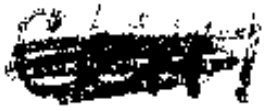
Original source details: GELFAND-P000218

Berezin
Kirillov' Grassmann analysis.

— • —

GELFAND-P000219

16/X



Uncertain reading. [Crossed out: illegible.]

Original source details: GELFAND-P000219

S. G. Krein.

On interpolation of linear operators.

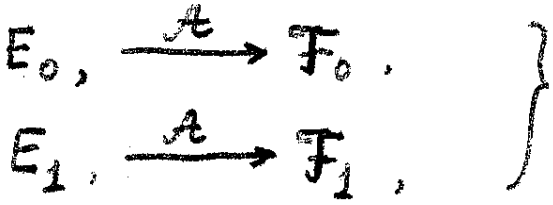
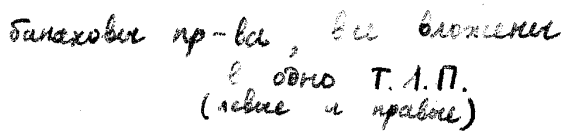


Diagram and typed description.

$$\left.\begin{array}{l} E_0, \xrightarrow{A} F_0, \\ E_1, \xrightarrow{A} F_1, \end{array}\right\}$$ Banach spaces,

Original source details: GELFAND-P000219

all are embedded into one T. L. S. (the left and the right ones)



Original source details: GELFAND-P000219

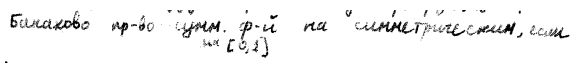
Manuscript notation. The last word reads вложены (“are embedded”).

A is bounded.

E, F — an interpolation pair if, for $\forall$ A, boundedness at 0 and 1 implies boundedness of $A : E \longrightarrow F$.

All spaces will be spaces of summable functions.

A Banach space of summable functions [Inserted below the line: on $[0, 1]$.] is called symmetric if



Original source details: GELFAND-P000219

Manuscript notation. The main line has the abbreviation на (“is called”). A separate insertion below reads “on $[0, 1]$”.

1) $y(t) \in E$

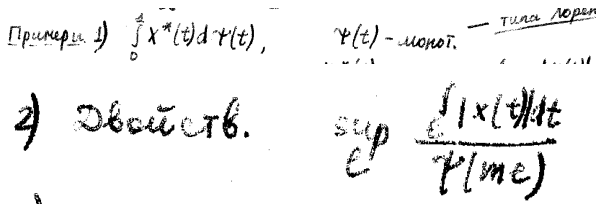
$$|x(t)| \le |y(t)| \quad \Longrightarrow \quad x \in E \qquad \text{and} \qquad \|x\| \le \|y\|$$

2) The norm does not change under transformations preserving measure (one may consider an equimeasurable monotone one)

Then $L_\infty \subset E \subset L_1$.

Examples. 1)

$$\int_0^1 x^*(t)\, d\psi(t), \qquad \psi(t) \text{ — monot.} \qquad \text{— of Lorentz type}$$



Original source details: GELFAND-P000219

Uncertain reading. The same three-armed sign occurs in the differential, the monotonicity condition and the denominator. The provisional reading is ψ. An alternative is Ψ.

$x^*(t)$ — a rearrangement of $|x(t)|$

2) Dual. $\sup\limits_e \dfrac{\int_e |x(t)|\, dt}{\psi(me)}$

3) and Orlicz spaces (these include L_p)

[At the left above the line:] 1st step.

Riesz–Thorin theorem.

GELFAND-P000220

2nd step.

Marcinkiewicz theorem.

(Less is required at the endpoints)

$$n_x(\tau) = \text{mes}\{t : |x(t)| \ge \tau\}$$

$$[n_{Ax}(\tau)]^{1/q_i} \le \frac{C}{\tau} \|x\|_{L_{p_i}}$$

— a less restrictive condition.

Most operators of analysis satisfy the Marcinkiewicz conditions at the endpoints

Matsaev and Dikarev — more precisely, from L_p into a space narrower than L_q (of Lorentz type)! — and it is already minimal (in the middle)

(Minimal for all operators under conditions both of Marcinkiewicz and of Riesz–Thorin) —

— this is for improving operators:

$$\begin{matrix} q_1 > p_1 \\ q_2 > p_2 \end{matrix} \quad \Big\| \quad \text{— as in the Marcinkiewicz theorem.}$$

$p = q$ — maps into itself — then all the other scales do too.

Let $\begin{matrix} L_1 \longrightarrow L_1 \\ L_\infty \longrightarrow L_\infty \end{matrix}$ and $\|A\| \le 1$

List all [Crossed out: illegible.] spaces for which $\|A\| \le 1$ as well.



Uncertain reading. [Crossed out: illegible.]

Original source details: GELFAND-P000220

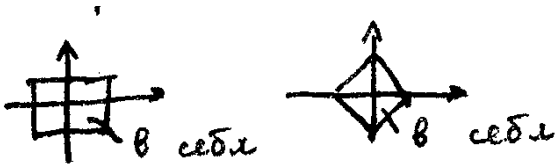
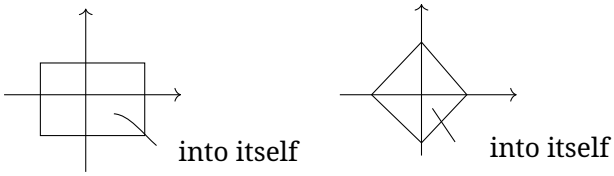


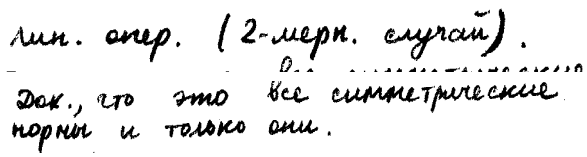
Diagram and typed description. The extra stroke curves at the rectangle and runs obliquely at the diamond. Their directions remain uncertain.



Original source details: GELFAND-P000220

[To the right of the figures:] linear oper. (2-dimensional case).

Док. [prove / proved], that these are all symmetric norms and only they.



Uncertain reading. The abbreviation “Док.” is readable; the full verb form is not written. Possible expansions: prove / proved.

Original source details: GELFAND-P000220

General theorem — B. S. Mityagin.

It is necessary and sufficient that the space be symmetric.

(also in the ∞-dimensional case)

$L_p \xrightarrow{A} L_p$ what then?

$p \neq 1, \infty$

(Marcinkiewicz case)

$$\| \underset{\substack{\text{char. fn. } e \\ \uparrow}}{\chi_e} \|_E = \varphi(me).$$

(symmetry is again necessary)

the resulting function $\varphi(t)$ — a characteristic of the space E.

$$\ell_E = \underline{\lim}_{t\to 0}\frac{\varphi(2t)}{\varphi(t)} \qquad u_E = \overline{\lim}_{t\to 0}\frac{\varphi(2t)}{\varphi(t)} \qquad 1 \le \ell_E \le u_E \le 2$$

GELFAND-P000221

[source number: 9]

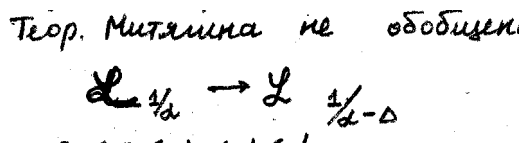
For L_p they are both equal to $2^{1/p}$

Boundedness of all such A in $E \Longleftrightarrow 1 < \ell_E \le u_E < 2$

(Semyonov's 1st theorem)

Mityagin's theorem has not been generalized.

$$L_{1/\alpha} \longrightarrow L_{\frac{1}{\alpha-\Delta}}$$



Original source details: GELFAND-P000221

Uncertain reading. The minus sign and Δ sit on the lower level with α after the slash. The preferred reading is $1/(\alpha - \Delta)$. The manuscript has no parentheses.

$0 < \Delta \le \alpha_0 < \alpha < \alpha_1$.

How can one list the spaces in which operators of this type are bounded?

$$2^{\alpha_0} < \ell_E \le u_E < 2^{\alpha_1} \quad \Longrightarrow \quad E \longrightarrow E_\Delta,$$

where



Original source details: GELFAND-P000221

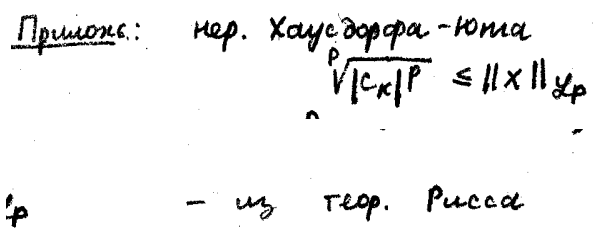
Uncertain reading. After "where" there are two dense short signs; their exact reading and whether they are crossed out are undetermined.

$$\|x\|_{E_\Delta} = \|x^*(t)t^{-\Delta}\|_E$$

(This generalizes the theorem of Dikarev and Matsaev — the best space)

Applic.: Hausdorff–Young inequality

$$\sqrt[p]{|c_k|^p} \le \|x\|_{L_p} \qquad \text{— from Riesz's theorem}$$



Original source details: GELFAND-P000221

Manuscript notation. The first radicand reads $|c_k|^p$. A summation sign is visible only under the next radical.

Poia inequality:

$$\sqrt[p]{\sum k^{p-2}c_k^{*p}} \le C\|x\|_{L_p}$$

c_k^* — the monotone rearrangement of the $|\ |$ coefficients.

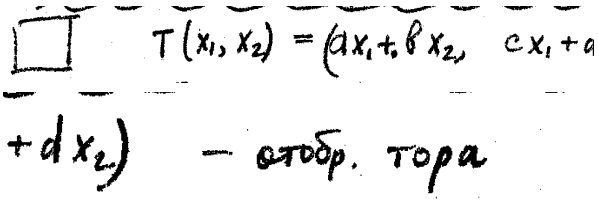
For monotone decreasing Fourier coefficients — two-sided inequalities (Hardy–Littlewood)

Between L_1 and $L_{1,5}$ there is no Hilbert interpolation space (Semyonov)

Gelfand's question: an abstract description of interpolation spaces. (it is unknown whether they are spaces of functions.)

Ya. Sinai.

Construction of invar. measures of smooth maps Markov partitions



Original source details: GELFAND-P000221

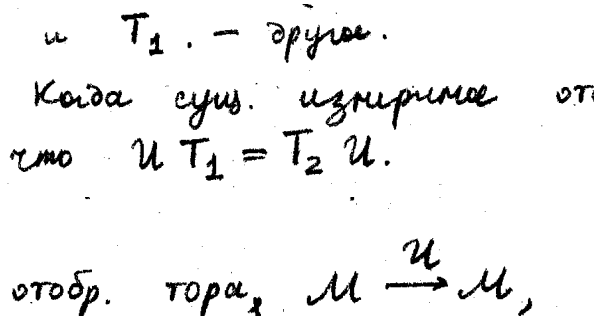
Diagram and typed description.



$T(x_1, x_2) = (ax_1 + bx_2,\ cx_1 + dx_2)$ — a map of the torus

and T_1. — another.

When does there exist a measurable map of the torus, $M \xrightarrow{U} M$, such that $UT_1 = T_2U$.



Original source details: GELFAND-P000221

Uncertain reading. The description has T and T_1. The equation has T_1 and T_2. The looped glyph above the arrow recurs on both sides of the equation. The provisional reading is U.

Irrational eigenvalues $\Rightarrow$ real $\Rightarrow$ ergodic, the spectrum is Lebesgue of countable multiplicity.

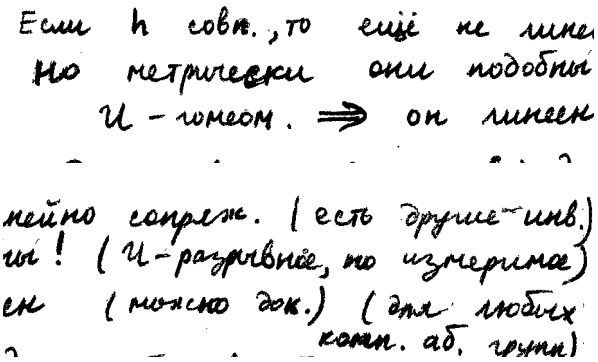
Entropy $h(T) = \ln \lambda \quad (\lambda > 1)$

GELFAND-P000222

If h agrees, this does not yet give linear conjugacy (there are other-inv.)

But metrically they are similar! (U — discontinuous, but measurable)

U — a homeom. $\Longrightarrow$ it is linear (one can prove this) (for any compact abelian groups)



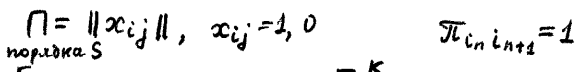
Original source details: GELFAND-P000222

Uncertain reading. The looped map sign in both lines: U / u. Candidate: U; letter case is undetermined.

Ω_{II} — the space of seq. $\omega = \{i_n\}$ — sequences infinite in both directions.

$i_n \in \{1, \ldots, S\}$

$$\underset{\text{of order } S}{\Pi} = \|x_{ij}\|, \quad x_{ij} = 1, 0 \qquad \pi_{i_n i_{n+1}} = 1$$



Original source details: GELFAND-P000222

Manuscript notation. The matrix definition writes x_{ij}. The admissibility condition writes $\pi_{i_n i_{n+1}}$.

(only such sequences ω are admissible.)

We shall assume that Π^k consists of positive entries.

Consider the shift (the shift transformation on Ω).

Example. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ — does not satisfy the cond., but Ω — sequences only of 1 or [Crossed out] only of 2



Original source details: GELFAND-P000222

Uncertain reading. A densely crossed-out sign follows “or”. Its reading is undetermined.

$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ — anything may be put in

$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ — look into it.

on Ω_Π we would like to specify an invariant measure μ with max entropy:

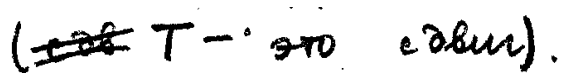


Original source details: GELFAND-P000222

Uncertain reading. [An illegible crossed-out entry to the left of the formula.]

$$h = \max_{\mu} h_\mu(T)$$

([Crossed out] T — this is the shift).

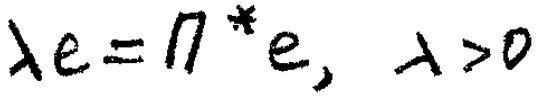


Original source details: GELFAND-P000222

Uncertain reading. A looped ending is visible beneath two diagonal cancellation strokes. The readings “where” and “shift” remain tentative.

One must consider e with positive coordinates. $\lambda e = \Pi^* e$, $\lambda > 0$

Manuscript notation. The star is at the upper right of Π.



Original source details: GELFAND-P000222

(unique.)

(G. If all entries and all minors of order 2 are > 0, then there exist 2 eigenvectors, the 1st with positive entries, the 2nd — with the 1st change of sign —

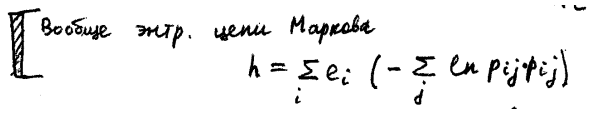
— the matrix of minors of order 2 — a representation of the matrix itself in the exterior product.)

Consider a Markov chain with probabilities $p_{ij} = \dfrac{e_j \pi_{ij}}{\lambda e_i}$ —

[At the right:] — entropy — $\ln \lambda$

In general, the entropy of a Markov chain

$$h = \sum_i e_i \left(- \sum_j \ln p_{ij} \cdot p_{ij} \right)$$



Manuscript notation. The bracket at left is hatched. The outer sum has e_i. A dot separates $\ln p_{ij}$ from p_{ij}.

Original source details: GELFAND-P000222

the measure of max entropy is unique.

The measure in Ω_Π on the Σ-alg. generated by cylindrical sets.

GELFAND-P000223

[source number: 10]



Diagram and typed description. Page number 10 with a line at its left and below it.

10

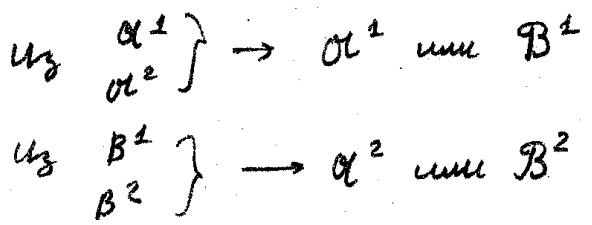
Original source details: GELFAND-P000223

a) $\{a_1^1, \ldots, a_p^1;\ a_1^2, \ldots, a_q^2;\ b_1^1, \ldots, b_r^1;\ b_1^2, \ldots, b_s^2\}$

Π

$$\text{from } \left.\begin{matrix}\alpha^1 \\ \alpha^2\end{matrix}\right\} \longrightarrow \alpha^1 \text{ or } B^1$$

$$\text{from } \left.\begin{matrix}\beta^1 \\ \beta^2\end{matrix}\right\} \longrightarrow \alpha^2 \text{ or } B^2$$



Uncertain reading. Both transitions have an uppercase B with an upper loop on the right. The left-hand forms resemble β on the second line and α on the first. Case on the left is uncertain.

Original source details: GELFAND-P000223

[In the right margin:] ‖ a partition is constructed from an integer matrix of order 2.

$$\lambda^2 - (p+s)\lambda + (ps - qr) = 0, \qquad \boxed{ps - qr = 1}$$

condition.

b) Ω_{Π_1} $\pi_{ij} = 1$ and $\pi_{N1} = 0$ (only in the 1st position)

Prop. The shifts in a) and b) are isomorphic.

⌈Everything now reduces to proving that
an algebraic automorphism ⇒
⇒ a partition.

Proof of the proposition.

Alphabet b): $A_1, \dots, A_p, \quad B_1, \dots, B_s$.

(Isomorphism of any 2 Ω_Π with the same entropy).

Different groups are encoded (using the symbols A_i, B_i).

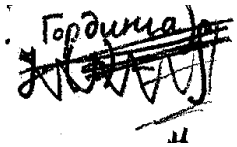
The space with the 1st max measure passes into a space with another max measure.

Next time: Markov partitions and how invariant measures are constructed with their help.

$23/\overline{\text{X}}$. Leray. Singularities of Feynman integrals.

N. Nilsson, Arkiv för Math. (Swede) 1964–1965

(student of Gårding)



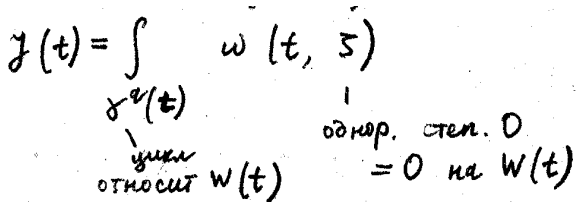
Original source details: GELFAND-P000223

Undeciphered passage. A short formula below "(student of Gårding)" is crossed out. Possible beginning: $J(\dots)$; the remaining symbols are illegible.



Original source details: GELFAND-P000223

Uncertain reading. A separate crossed-out sign below the formula: J / γ; its reading is undetermined.



Original source details: GELFAND-P000223

Uncertain reading. The cycle superscript has a loop and descender, favouring q. The same angular sign occurs in the argument of ω and in the coordinate line. The provisional reading is ξ. An alternative is ζ.

$$J(t) = \int_{\gamma^q(t)} \omega(t, \xi)$$

cycle relative to $W(t)$ — homog. deg. 0 $= 0$ on $W(t)$

$t \in T = \mathbb{C}^n$,

$\xi \in \mathbb{C}^{\ell+1}$

$\omega(t, \xi)$ on $\mathbb{P}^\ell$, since it is homogeneous; the cycle is there too.

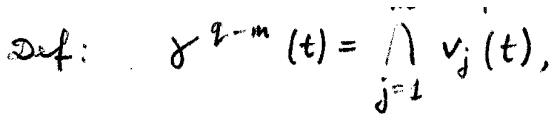
GELFAND-P000224

$W(t) = \bigcup_k W_k(t)$ — hyperplanes, algebraic functions of t, singularities may occur (not in general position)

$t \in S_0[W] \iff$ the dimension of the intersection of any number changes there.

Def:

$$\gamma^{q-m}(t) = \bigcap_{j=1}^{m} V_j(t),$$



Manuscript notation. The large sign has a curved top and descending sides. It is set as $\bigcap$.

Original source details: GELFAND-P000224

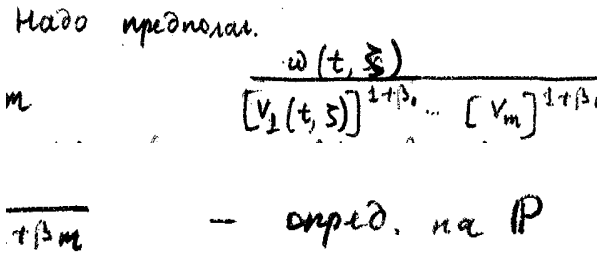
$$V_j(t) = v_j(t, x) = 0$$

)

algebraic set

One must assume

$$\frac{\omega(t,\xi)}{[v_1(t,\xi)]^{1+\beta_1} \dots [v_m]^{1+\beta_m}} \quad \text{— defined on } \mathbb{P}$$

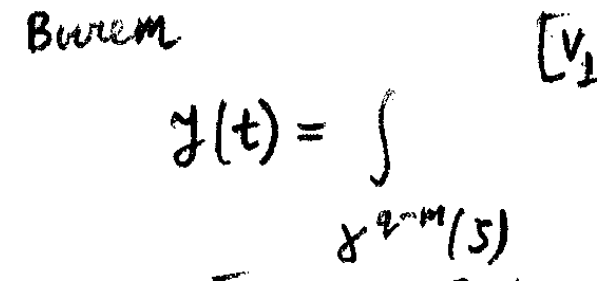


Uncertain reading. The coordinate signs in the argument of ω and first denominator: ξ / s; displayed candidate: ξ. The last factor is written $[v_m]^{1+\beta_m}$.

Original source details: GELFAND-P000224

[On the left:] Residue

$$J(t) = \int_{\gamma^{q-m}(s)} \frac{d^{\beta_1+\dots+\beta_m}\omega}{[dv_1]^{1+\beta_1} \wedge \dots \wedge [dv_m]^{1+\beta_m}}$$



Uncertain reading. The sign in parentheses has a short top stroke and a lower curve. The reading s is provisional. An alternative is ξ.

Original source details: GELFAND-P000224

The singular support is also described in the general case

(On the covering space of the complement the form will be holomorphic)

Thm. Then $J(t)$ has algebraic singular support.



Uncertain reading. Right-margin note: “of type st. algebraic” / “of type sm. algebraic”; the first word and middle abbreviation are not uniquely established.

Original source details: GELFAND-P000224

(i.e. in T there exists an algebraic set such that $J(t)$ is holomorphic on a covering of the complement.)

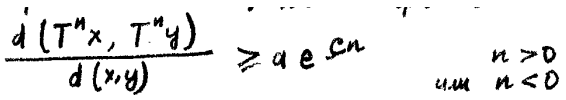
Leray described the singular support in the case of planes.

$30/\underline{\text{X}}$ Sinai. U-diffeom.

characteristic exp growth of the distance between nearby points (separation)

either as $t \longrightarrow +\infty$, or as $t \longrightarrow -\infty$.

$$\frac{d(T^n x, T^n y)}{d(x,y)} \geq ae^{cn} \qquad \begin{matrix} n>0 \\ \text{or } n<0 \end{matrix} \qquad (x, y \text{ — nearby})$$



Manuscript notation. The exponent reads cn. The two lines at the right give $n > 0$ and "or $n < 0$".

Original source details: GELFAND-P000224

Consider the points $d(T^n y, T^n x) \leq ae^{-cn} d(y, x)$,

Suppose that $\forall\, x$ this is a k-dimensional subset.

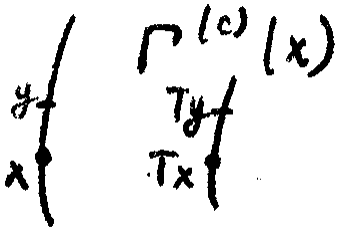
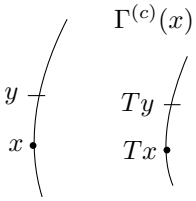


Diagram and typed description.



Original source details: GELFAND-P000224

This is a contracting subset (hence the index $\underline{\underline{c}}$)

$$d(T^n y', T^n x) \leq ae^{+cn} d(y', x) \qquad (n \longrightarrow -\infty).$$

Similarly we obtain $\Gamma^{(p)}(x)$ — an expanding subset.

(ℓ-dimensional).

Then the choice does not depend on a and c.

GELFAND-P000225

[source number: 11]



Diagram and typed description. Page number 11 with a line above, left, and below.



Original source details: GELFAND-P000225

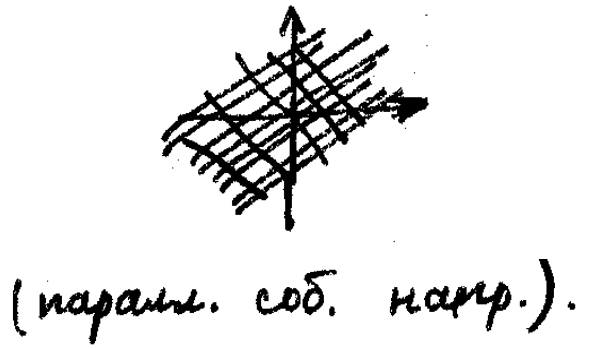
U-system: $k + \ell = n$ and the tangent spaces to the submanifolds are uniformly transverse

1. $k + \ell = n$
2. transversality.
3. Smoothness (not written out explicitly).

We obtain 2 foliations $\gamma^{(c)}$ and $\gamma^{(p)}$.

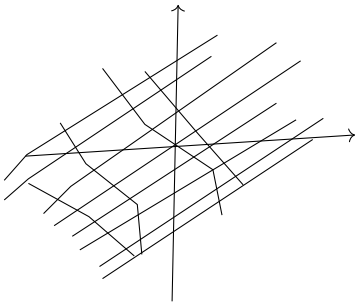
Example: linear transformation of the torus $(x, y) \longrightarrow (ax + by, cx + dy)$.

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = 1.$$



Original source details: GELFAND-P000225

Uncertain reading. Two families of inclined lines and axes with arrows upwards and rightwards. The number of coinciding lines and continuations within dense crossings are not uniquely established. Individual lines have no labels.



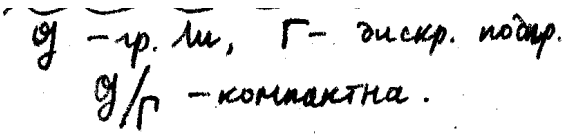
(parallel to the eigendirections).

A nonlinear perturbation (periodic and smooth) is also a U-diffeomorphism and the rotation number is the same.

Method of constructing U-diffeomorphisms.

g — a Lie group, Γ — a discrete subgroup.

g/Γ — compact.



Original source details: GELFAND-P000225

Manuscript notation. The group sign has a loop and descender. The uppercase G below denotes the Lie algebra.

$\underbrace{T_{\text{alg}}}$ — an algebraic automorphism of g, leaving Γ fixed $\quad (T_{\text{alg}}\Gamma = \Gamma)$

automorphism as a group

Then it acts on g/Γ.

Consider G — the Lie algebra; $\hat{T}_{\text{alg}}$ — the corresponding automorphism of G; if its

[At the left edge of the following line:] eigenvalues.

Among them there are no [Crossed out] eigenvalues $= 1$ in mod $\Longleftrightarrow$

$\Longleftrightarrow$ this is a U-diffeomorphism.



Original source details: GELFAND-P000225

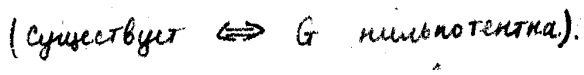
Uncertain reading. Crossed out: $\Longleftrightarrow$ / $\Longrightarrow$; endpoint directions are not established.



Original source details: GELFAND-P000225

Manuscript notation. After mod there are two horizontal strokes and two arrowheads. The sign recurs at the start of the next line.

(It exists $\Longleftrightarrow$ G is nilpotent).



Original source details: GELFAND-P000225

Manuscript notation. The first word reads Существует (“It exists”). The letter before the final т is е.

Further. A small nonlinear perturbation $T = T_{\text{alg}} + \Delta T$ is also a U-diffeomorphism.

Main problem: topological similarity to the algebraic one.

For small perturbations — from Anosov’s general theorem

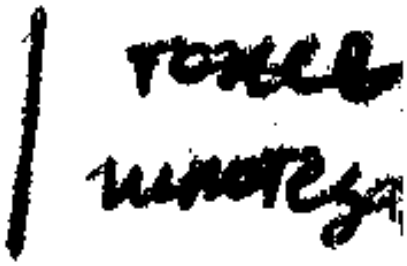
(A small perturbation of a U-diffeomorphism is topologically similar to the original).

Always.

$$M = M_1 \cup M_2$$

M_2 — wandering points

M_1 — nonwandering points. — the condition U holds on them



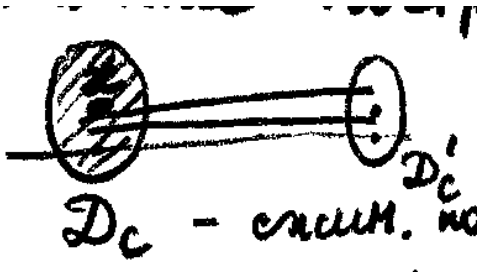
Original source details: GELFAND-P000225

Diagram and typed description. A vertical rule separates the right-margin note.

> also a
> conjecture

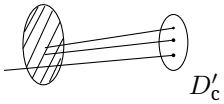
GELFAND-P000226

An important construction:



Original source details: GELFAND-P000226

Diagram and typed description.

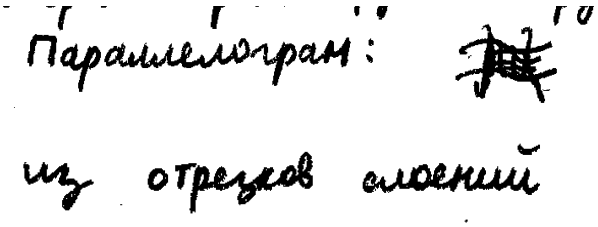


natural isomorphism of our pieces of contracting leaves (along noncontracting leaves).

D_c — a contracting subset, dim $= k$.

(π — the canonical isomorphism.)

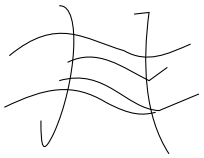
π is not always smooth (!), but it is absolutely continuous, i.e. the natural measures pass into one another.



Original source details: GELFAND-P000226

Uncertain reading. Two nearly vertical curved strokes and transverse curves; the number of coinciding strokes and their continuations at the dense middle crossings are not uniquely established. No labels or arrows are visible.

A parallelogram:



made of segments of foliations

(all its leaves are canonically isomorphic: both contracting and expanding.)

$$\Gamma(U) = \Gamma_p(U) \cup \Gamma_c(U)$$

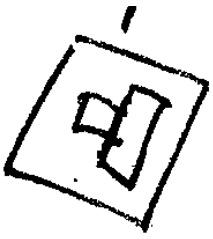
Suppose there is a system of parallelograms U_i adjoining one another.

$$M = \bigcup \overline{U}_i$$
$$U_i \cap U_j \subset \Gamma(U_i) \cap \Gamma(U_j) \qquad (\Gamma \text{ — boundaries}).$$

α — such a partition into parallelograms. Consider the boundaries:

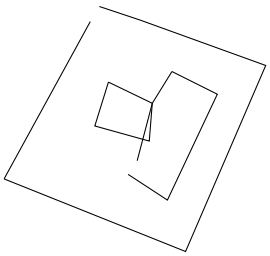
$$\Gamma_p(\alpha) = \bigcup_i \Gamma_p(U_i)$$
$$\Gamma_c(\alpha) = \bigcup_i \Gamma_c(U_i)$$

[Drawing on the right:]



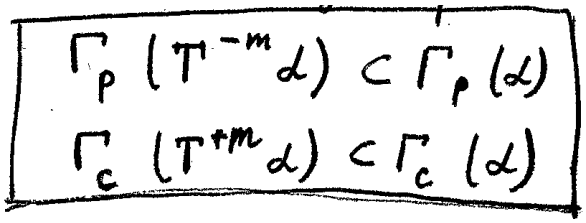
Original source details: GELFAND-P000226

Diagram and typed description.



This α is called Markov if there exists an m such that

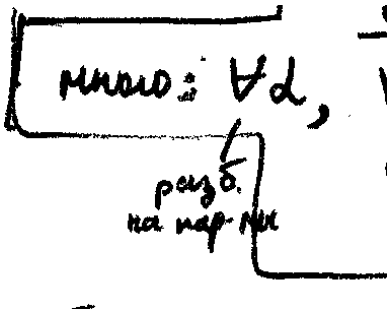
$$\boxed{\begin{array}{l}\Gamma_{\mathrm{p}}(T^{-m}\alpha) \subset \Gamma_{\mathrm{p}}(\alpha) \\ \Gamma_{\mathrm{c}}(T^{+m}\alpha) \subset \Gamma_{\mathrm{c}}(\alpha)\end{array}}$$



Manuscript notation. The upper exponent is $-m$ and the lower one is $+m$. One box surrounds both lines.

Original source details: GELFAND-P000226

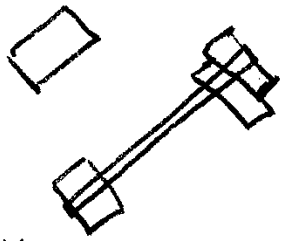
Thm. (technical) Markov partitions exist and there are many of them: $\forall$ α (partition into parallelograms), $\forall \varepsilon > 0$ $\exists m$ and a partition $\widetilde{\alpha}$, such that it will be at a distance of less than ε from the original.



Manuscript notation. The pointer links the note “partition into parallelograms” to $\forall\alpha$.

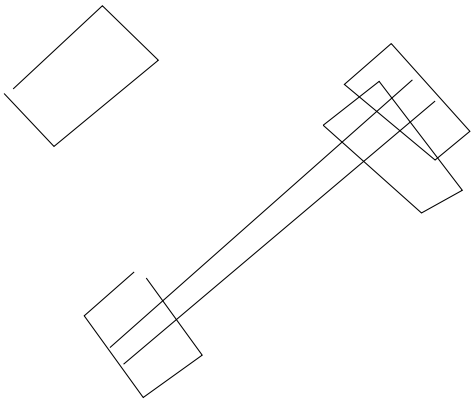
Original source details: GELFAND-P000226

[On the left:]



Uncertain reading. A separate inclined quadrilateral and a narrow inclined strip with transverse quadrilaterals at its ends. The continuations of several superposed contours at the upper right end are not uniquely established. There are no arrows.

Original source details: GELFAND-P000226



the ends cannot be in the middle.
(original parallelogram.)

— • —

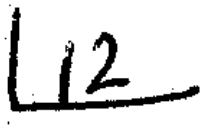
If α is a partition with m, then one can make $m = 1$, taking the intersection

$$\alpha_1 = \alpha \cdot T^{-1}\alpha \cdot \ldots \cdot T^{-m+1}\alpha \text{ —}$$

— Markov with $m = 1$

GELFAND-P000227

[source number: 12]



Original source details: GELFAND-P000227

Diagram and typed description. Page number 12 with a line at its left and below it.

12

— from

$$\Gamma_{\mathrm{p}}(\alpha_1) = \Gamma_{\mathrm{p}}(\alpha) \cup \Gamma_{\mathrm{p}}(T^{-1}\alpha) \cup \cdots \cup \Gamma_{\mathrm{p}}(T^{-m+1}\alpha)$$

Symbolic dynamics for a U-diffeomorphism.

(Suppose there is a Markov partition and with $m = 1$.)

$$\pi_{j_1 j_2} = \begin{cases} 1 & V_{j_1} \cap TV_{j_2} \neq \varnothing \\ 0 & \text{in the remaining cases} \end{cases}$$

Π

Ω_Π — the space of points

$$\omega = \{\ldots\ j_{-n}, \ldots, j_0, \ldots, j_n, \ldots\}$$
$$\pi_{j_n j_{n+1}} = 1.$$

Now associate

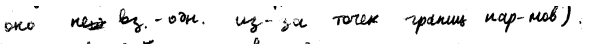
$$\omega \longrightarrow \bigcap_{-\infty}^{+\infty} T^n \overline{V}_{j_n} = x \in M \qquad (!)$$

(nonempty by the Markov property and, for a fine partition, consists of 1 point.)

We obtain a map $\varphi : \Omega_\Pi \longrightarrow M$

[A parenthesis begun on the right:] (from nonuniqueness.

it is not [Crossed out] one-to-one because of the points — boundaries of parallelograms).



Original source details: GELFAND-P000227

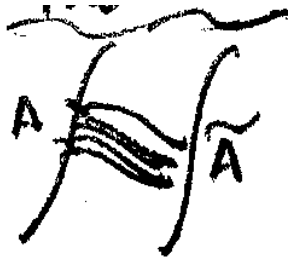
Uncertain reading. After the readable “not” a short entry is crossed out; possible “one-to-” is not established.

On a set of the 2nd category it is one-to-one.

Condition: Π^K consists of positive entries for some K.

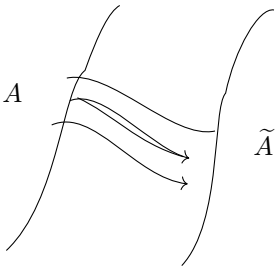
(i.e. all leaves $\Gamma^{(\mathrm{c})}$ and $\Gamma^{(\mathrm{p})}$ are dense in M — this is necessary and sufficient.)

On each leaf $\Gamma^{(c)}$ a σ-finite measure λ_c is introduced,
where
[On the left:]

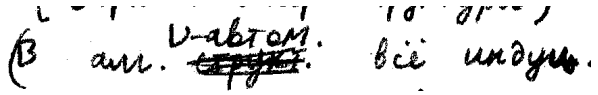


Original source details: GELFAND-P000227

Uncertain reading. The lines labelled $A, \widetilde{A}$ and connecting curves. Directed thickenings are visible at two lower right ends; the number of coinciding lower curves and character of the other ends are not uniquely established.



1) $\lambda_c(A) = \lambda_c(\widetilde{A})$
(measures of canonically isomorphic sets are equal.)
2) $\lambda_c(TA) = h\lambda_c(A) \quad (h < 1)$
(h — an algebraic number.)
— additional information.
(Reflects the homogeneous structure.)
(In an algebraic



Original source details: GELFAND-P000227

Uncertain reading. A word is crossed out, possibly “case”; its reading is not established.

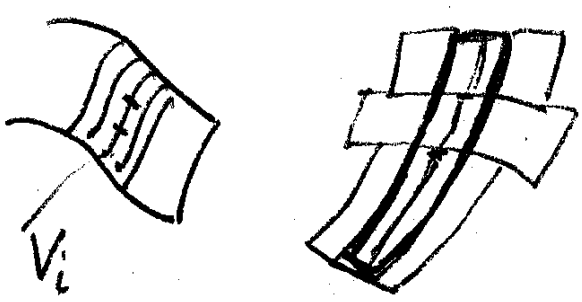
[Inserted above:] U-automorphism everything is induced by Haar measure.)
Corollary: there is an invariant measure, T is [Crossed out] a finite Markov chain.



Original source details: GELFAND-P000227

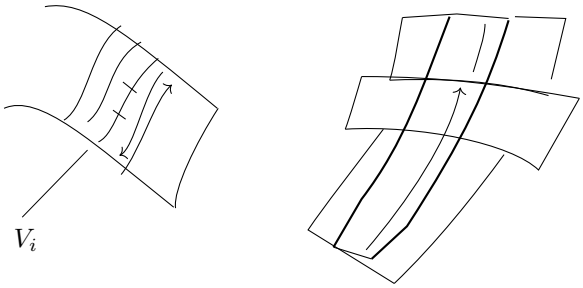
Uncertain reading. Crossed-out sign: Cyrillic “б” / “в”; its reading is not established.

Construction of this measure: $(n = 2)$

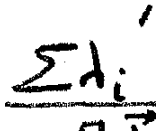


Original source details: GELFAND-P000227

Uncertain reading. Two diagrams with V_i, leader stroke, transverse marks, and visible directed strokes. Connections of thick lines at the base of the right diagram and some endpoint directions remain undetermined. There is no arrow between the diagrams.



$$\lambda_i h^{-1} = \sum \lambda'_i = \sum \pi_{ii'} \lambda_{i'}, \quad \text{i.e.}$$



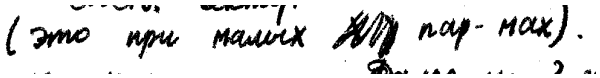
Original source details: GELFAND-P000227

Manuscript notation. In the first sum the prime sits high above the subscript. In the second sum both primes are at subscript height.

$$\boxed{h^{-1}\vec{\lambda} = \Pi\vec{\lambda}.}$$

— eigenvector.

(This is for [small?] [Crossed out] parallelograms).



Original source details: GELFAND-P000227

Uncertain reading. The letterforms favour “малых” (small), as on manuscript page 11. “Наших” (our) remains possible. The next word is crossed out.



Original source details: GELFAND-P000227

Uncertain reading. A short crossed-out entry is illegible.

The values of the measure on a segment are given. Then from the 2nd property.

[Below the last line:] (on a smaller segment.)

GELFAND-P000228

Entropy



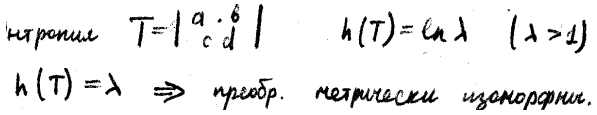
Original source details: GELFAND-P000228

Uncertain reading. The final stroke favours Энтропия (Entropy). The ending remains provisional.

$$T = \begin{vmatrix} a & \cdot & b \\ c & & d \end{vmatrix}$$

$$h(T) = \ln \lambda \quad (\lambda > 1)$$

$$h(T) = \lambda$$

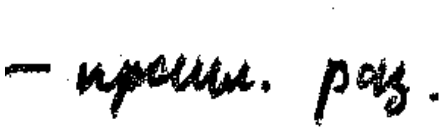


Original source details: GELFAND-P000228

Manuscript notation. The upper line writes $h(T) = \ln \lambda$. The next writes $h(T) = \lambda$. A dot stands between a and b.

$\Rightarrow$ the transformations are metrically isomorphic.

Ω_{Π_1} maps onto Ω_{Π_0} (stand.) —

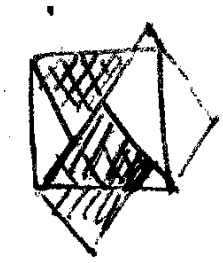


Original source details: GELFAND-P000228

Uncertain reading. The abbreviation before раз. provisionally reads прежн. (previous). The middle strokes merge.

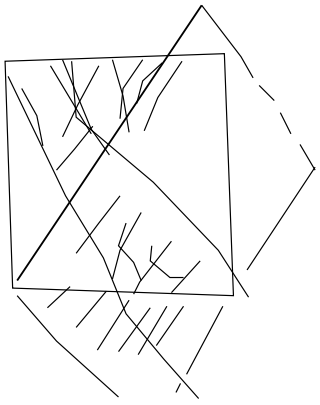
And it remains to map the torus onto Ω_{Π_1}.

Π_1 is of a certain type: 4 groups of letters and properties of transitions — see earlier.



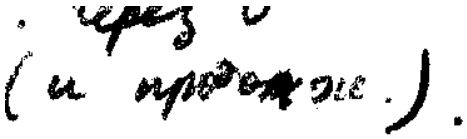
Original source details: GELFAND-P000228

Uncertain reading. A nearly rectangular frame, oblique edges projecting above, to the right and below, and hatching. Continuations of lines at dense crossings and exact boundaries of the hatched parts are not uniquely established. There are no vertex labels or arrows.



— a partition of the torus into 2 parallelograms (the sides $\parallel$ the eigendirections)

The sides lie on a line through 0 in an eigendirection (and its cont.?).

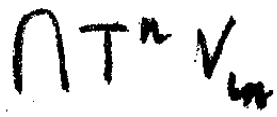


Original source details: GELFAND-P000228

Uncertain reading. The abbreviation after "and" appears to read продолж. (continuation). The middle letters overlap.

This is a Markov partition (the coarsest). It is because of it that one succeeds in solving the isomorphism problem.

$$\bigcap T^n V_{\omega_n}$$

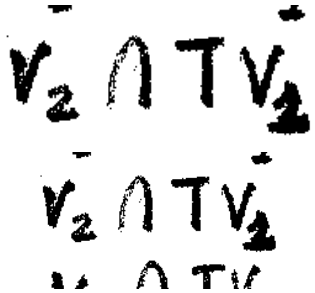


Original source details: GELFAND-P000228

Uncertain reading. The mark below V is looped. A separate n is hard to distinguish. V_{ω_n} is provisional.

may consist of a large number of points.
Consider first 4 parallelograms:

$$V_1 \cap TV_1$$
$$V_2 \cap TV_2$$
$$V_1 \cap TV_2$$
$$V_1 \cap TV_2$$



Original source details: GELFAND-P000228

Manuscript notation. The right subscript on the second line has a foot like the 2 in the following lines. The third and fourth lines both read $V_1 \cap TV_2$.

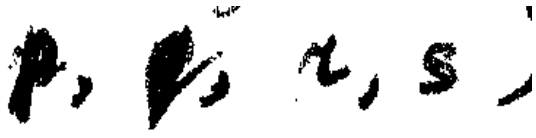
(



Original source details: GELFAND-P000228

Uncertain reading. Уже (Already); the word is readable, but whether a transverse stroke crosses it out is undetermined.

a disconnected partition. Form a partition — — its connected components. Their numbers, respectively — —



Original source details: GELFAND-P000228

Uncertain reading. p, q, r, s; the second sign is heavily retraced and q remains a candidate.

)

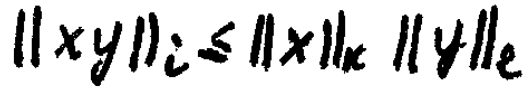
GELFAND-P000229

: Year 1968.

12/II Question (Gelfand)

R — a countably normed ring:

$$\|xy\|_i \le \|x\|_k \|y\|_e$$

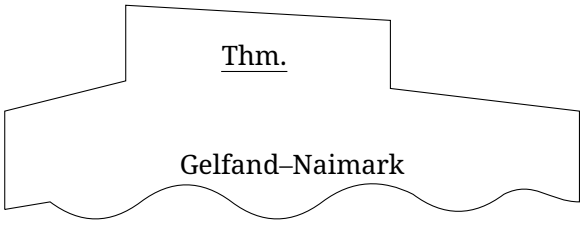


Uncertain reading. The last subscript has a low closed loop like e. The reading e is provisional. An alternative is ℓ.

Original source details: GELFAND-P000229



Diagram and typed description.



Original source details: GELFAND-P000229

for



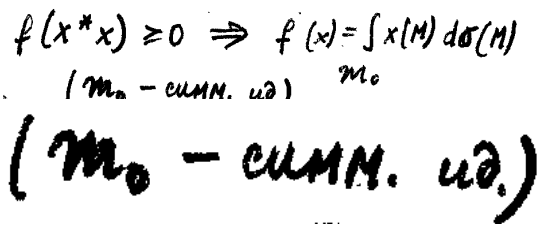
Uncertain reading. [Superscribed sign: superposed loops and strokes; its reading is undetermined.]

Original source details: GELFAND-P000229

normed rings:

$$f(x^*x) \ge 0 \Rightarrow f(x) = \int_{\mathcal{M}_0} x(M)\, d\sigma(M)$$

($\mathcal{M}_0$ — symm. ideals)



Uncertain reading. The slanted letter with a subscript zero occurs below the integral and in the parenthesis. It is provisionally set as $\mathcal{M}_0$.

Original source details: GELFAND-P000229

Extension — ? (Not always: one must characterize the class axiomatically).

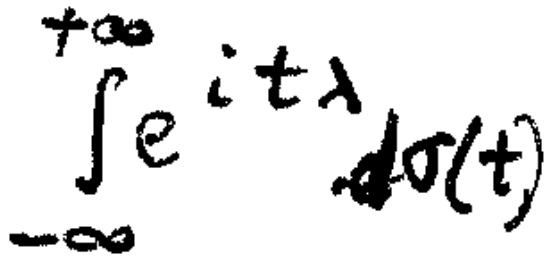
$$x^*(M) = \overline{x(M')}, \quad M = M'$$

— symm.

L^1 with convolution, $\overline{x(-t)} = x^*(t)$

⇒ Bochner's theorem

$$\int_{-\infty}^{+\infty} e^{it\lambda}\, d\sigma(t)$$



Manuscript notation. The bounds are $-\infty$ and $+\infty$. The exponent is $it\lambda$ and the differential is $d\sigma(t)$.

Original source details: GELFAND-P000229

I. Bernstein.



Original source details: GELFAND-P000229

Diagram and typed description.

Grothendieck topology.

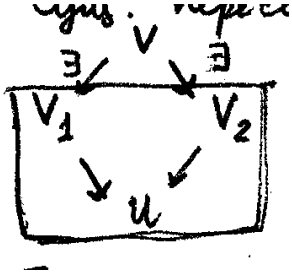
T is the following collection:

1) The category Cat T is specified (objects — open sets).

2) Certain families $\{\alpha_i : U_i \to U\}$ are specified, called coverings

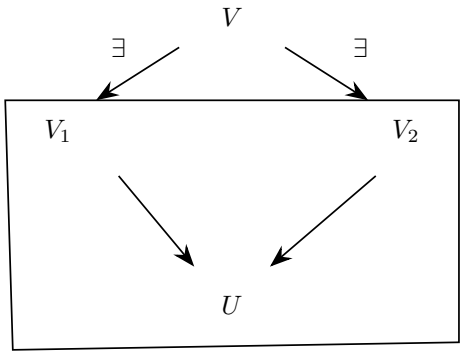
3) A system satisfying axioms is specified:

1. There exists an intersection — a product (fibered).



Original source details: GELFAND-P000229

Diagram and typed description.



(commut.).

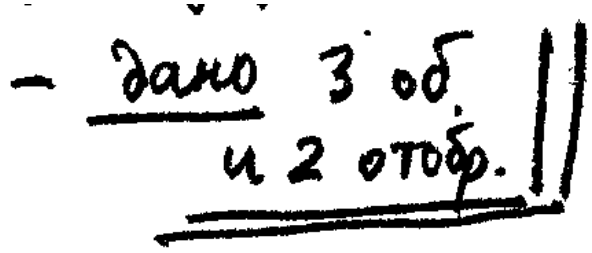


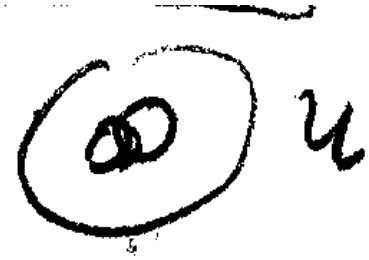
Diagram and typed description.

— given 3 objects

and 2 maps

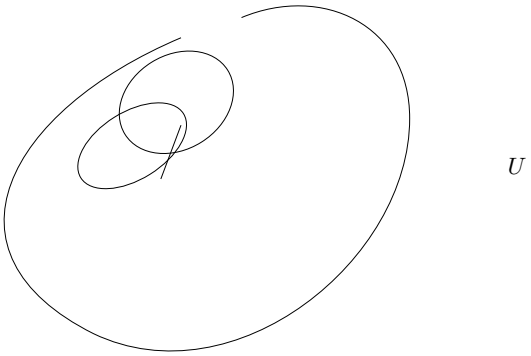
Original source details: GELFAND-P000229

$\boxed{V = V_1 \times_U V_2}$ — notation. Then there exists



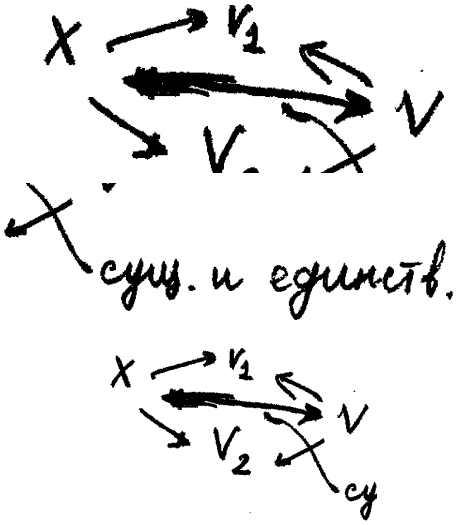
Original source details: GELFAND-P000229

Diagram and typed description.



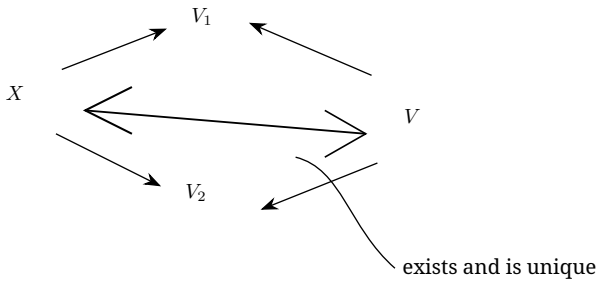
The universal property is required to hold:

$\forall X$, and $\forall$



Original source details: GELFAND-P000229

Uncertain reading. The central line has a clear arrowhead at right. The dense wedge at left may be a correction. The leader carries the note “exists and is unique”.



Depends essentially on U (in Grothendieck top.)

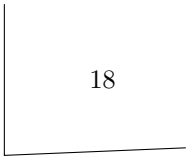
(If V exists, it is determined uniquely by the properties of the category)

GELFAND-P000230



Original source details: GELFAND-P000230

Uncertain reading. The upper right reads 18. The first digit is an upright stroke with no loop.



2. $\{U \xrightarrow{\text{id}} U\}$ — a covering.
3. $\{V_i \to V\}$ — a covering.

$\{V_{ij} \to V_i\}$ — a covering.

$\Rightarrow \{V_{ij} \to V\}$ — a covering.

4. $\{U_i \to U\}$ — a covering

$\{V \to U\}$ — a morphism

Then $\{U_i \times_U V \to V\}$

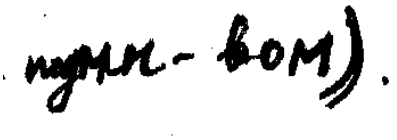
Example: unramified topology.

X — an ordinary topological space.

An open set: $U \to X$ — a local homeomorphism

(U — a topological space)

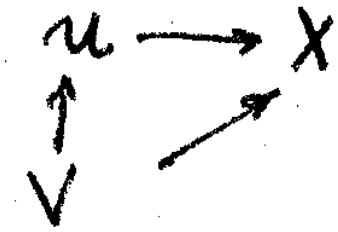
(a covering over X or its open subset).



Original source details: GELFAND-P000230

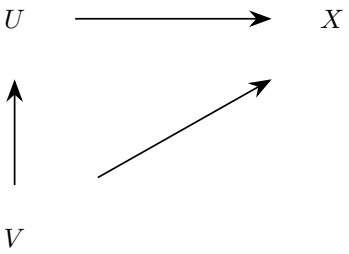
Manuscript notation. The word подмн-вом (subset) retains the opening под. No separate cancellation stroke is visible.

A map



Original source details: GELFAND-P000230

Diagram and typed description.



(any continuous one with a commutative diagram)

$\{U_i \to U\}$ — a covering, if the images



Original source details: GELFAND-P000230

Undeciphered passage. [The crossed-out word is undeciphered.]

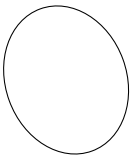
cover.

Axiom 1.



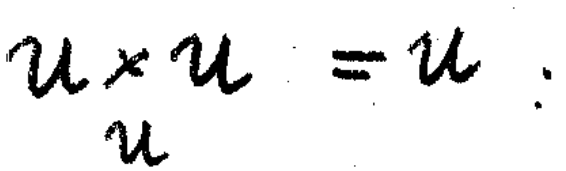
Original source details: GELFAND-P000230

Diagram and typed description.



U — a circle, U — also a circle, a map of degree 2 $U \to U$.

$$U \times_U U = U.$$



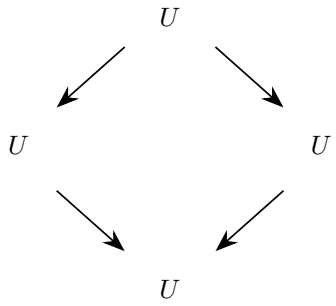
Original source details: GELFAND-P000230

Manuscript notation. The equality has one U after the equals sign.

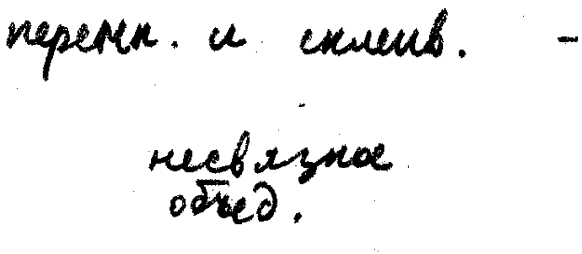


Original source details: GELFAND-P000230

Diagram and typed description.



$U_1 \times_X U_2$ — multipl. and gluing



Original source details: GELFAND-P000230

Manuscript notation. The first abbreviation is перемн. (multipl.), followed by и склеив. (and gluing).

— a disconnected union.

2 and 3 sheets $\Rightarrow$ 6 sheets

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \neq \mathbb{Z}_4,$$
$$\mathbb{Z}_2 \times \mathbb{Z}_3 = \mathbb{Z}_6.$$

GELFAND-P000231

Presheaf: • a functor

$$\text{Cat}\, T \to \text{Ab} \qquad \text{(contravar.)}$$
$$u \to P(u)$$

$$v \to u \;\Rightarrow\; P(u) \to P(v)$$

A sheaf — a presheaf satisfying the axioms:

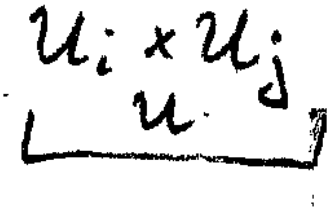
1. $\xi \in P(u)$

$\{u_i \to u\}$ — a covering

$$\xi \to \xi_i \qquad P(u) \to P(u_i)$$

ξ is uniquely determined by ξ_i

2. and is indeed determined, if ξ_i agree on intersections



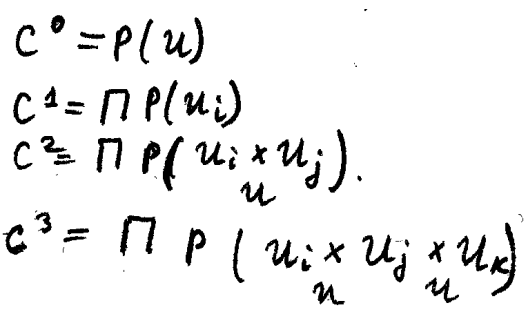
Original source details: GELFAND-P000231

Diagram and typed description.

$$u_i \times_u u_j$$

Čech cohomology is defined

$$C^0 = P(u)$$
$$C^1 = \prod P(u_i)$$
$$C^2 = \prod P(u_i \times_u u_j).$$
$$C^3 = \prod P(u_i \times_u u_j \times_u u_k)$$



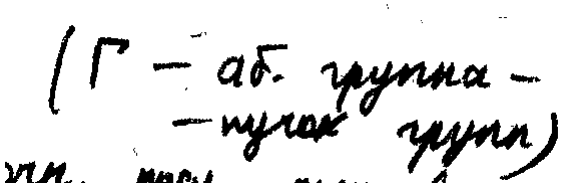
Original source details: GELFAND-P000231

Manuscript notation. The manuscript writes $C^0 = P(u)$ and begins the products at C^1. There are no indices under the product signs.

d is defined naturally

$\mathcal{F} \to \Gamma(X, \mathcal{F})$ — a functor.

(Γ — an abelian group —
— a sheaf of groups)



Original source details: GELFAND-P000231

Uncertain reading. Both lines of the parenthesis are retained. No separate symbol is visible before the dash in the second line; its implied subject is undetermined.

$$0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0.$$

— an exact sequence of sheaves.

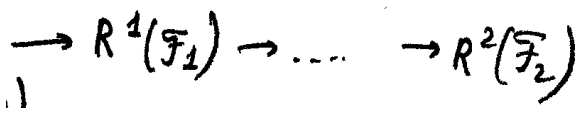
$$0 \to \Gamma(\mathcal{F}_1) \to \Gamma(\mathcal{F}_2) \to \Gamma(\mathcal{F}_3)$$

(left exact).

Derived functors exist. $R^i : S \to \text{Ab}$

S — arbit. sheaves (category).

$$\to R^1(\mathcal{F}_1) \to \ldots \to R^2(\mathcal{F}_2)$$



Manuscript notation. The ellipsis is followed by $R^2(\mathcal{F}_2)$. The opening arrow and the ellipsis are as written.

Original source details: GELFAND-P000231

(R^i — determined uniquely).

$$R^i(\mathcal{F}) \underset{\text{def}}{=} H^i(X, \mathcal{F}).$$

Finite group. Grothendieck top. from representations.

GELFAND-P000232

11/III Gelfand (and Kirillov)

$\mathfrak{g}$ — semisimple Lie algebra, (over C)

$U(\mathfrak{g})$ — its enveloping algebra

(polyn. in generators $e_1, \ldots, e_n$,

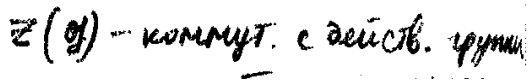
$$e_i e_k - e_k e_i = \sum C_{ik}^j e_j$$

$Z(\mathfrak{g})$ — centre of the enveloping algebra

(its elements — Laplace or Casimir operators).

Repr. of $G \Rightarrow$ diff. operators and commut.

$Z(\mathfrak{g})$ — commut. with the action of the group



Manuscript notation. The right edge of the line reads группы (group).

Original source details: GELFAND-P000232

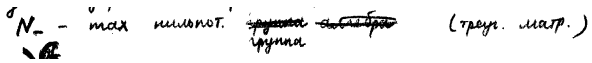
Such a repr. of the enveloping alg. by diff. operators is not necessarily faithful.

On G itself — faithful (transform.)

For $Z(\mathfrak{g})$ it makes no difference, left or right.

$\mathfrak{g}$ — semisimple Lie algebra. (over C).

N_- — max nilpot.



Uncertain reading. [crossed out: group algebra] [inserted below the first word: group]

Original source details: GELFAND-P000232

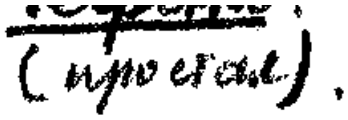
(triang. matr.)



Uncertain reading. Before the theorem, N_- and a diagonal stroke are visible. The upper right character is overwritten. The clearer notation in the theorem supports a provisional $N_- \backslash G$.

Original source details: GELFAND-P000232

Theorem: in $N_-\backslash G$ the repr. of $U(\mathfrak{g})$ is faithful.
(simple?).



Uncertain reading. The word below the heading probably reads простая (simple). Its last letters are faint.

Original source details: GELFAND-P000232

f on the group is regular if its translates — finite-dim. space
(sin on the line)
On a comp. group, functions are approx. by regular ones (completeness theorem)
Consider functions $f(g) \equiv f(n_- g)$ — invar. — ring.
Consider Spec of this ring. This is X.
$N_-\backslash G$ birat. equiv. to projective space.
Enveloping algebras are numerous, but division rings, of fractions are few

$$\underset{\substack{\text{division ring}\\ \text{of quotients}}}{D(\mathfrak{g})} \sim D_{n,k}$$

Nontriv. that embeds into a division ring (!) of quotients.

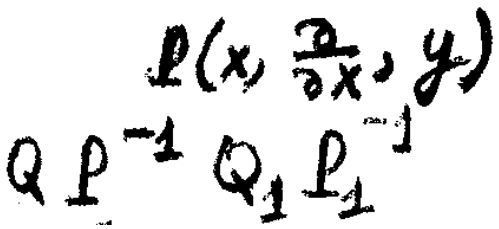
$$x_1, \dots, x_n; \quad \frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}, \quad y_1, \dots, y_k$$

$P(x, \frac{\partial}{\partial x}, y)$



Manuscript notation. The initial P has the loop and foot seen in the denominator letter on the next line.

Original source details: GELFAND-P000232



Manuscript notation. The letter recurs in P, P^{-1} and P_1^{-1}.

Original source details: GELFAND-P000232

$\underline{\underline{QP^{-1}}}$ $Q_1 P_1^{-1}$ and so on. Everything can be done in 2 steps.
(QP^{-1} suffices)
n — number of variables, k — number of parameters — invariants.

GELFAND-P000233

[source number: 23]



Diagram and typed description.

23

Original source details: GELFAND-P000233

For a semisimple group it is almost proved that $D(\mathfrak{g}) \sim D_{n,k}$.
(matr. is valid)



Manuscript notation. The words below the formula read матр. (matr.) and верна (is valid).

Original source details: GELFAND-P000233

For nilpotent proved, and probably for algebraic solvable.



Diagram and typed description.

Adjoin to $Z(\mathfrak{g})$ (K) operators — virtual Laplace operators

Original source details: GELFAND-P000233

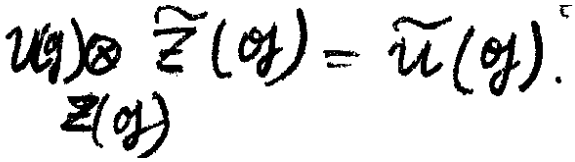
New ring $\tilde{U}(\mathfrak{g})$ — with adjoined left translations



Undeciphered passage. A short word or abbreviation is crossed out; its letters beneath the heavy strokes have not been established.

Original source details: GELFAND-P000233

$$U(\mathfrak{g}) \otimes_{Z(\mathfrak{g})} \tilde{Z}(\mathfrak{g}) = \tilde{U}(\mathfrak{g}).$$

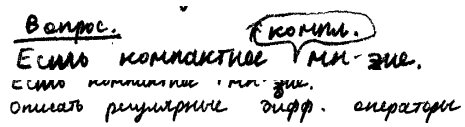


Manuscript notation. The manuscript puts $Z(\mathfrak{g})$ under $\otimes$ and writes tildes over the second factor and the U at right.

Original source details: GELFAND-P000233

$\tilde{Z}$ — Z with adjoined virt. Laplace operators.
(this is an algebraic extens., Galois group — Weyl group).
Theorem. The division ring of fractions of $\tilde{U}(\mathfrak{g})$ is $D_{n,k}$.
(For U unknown; done for unimodular).
Alg. extens. of $D_{n,k}$ — ?.

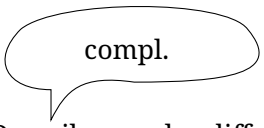
In the commut. case, alg. varieties.



Original source details: GELFAND-P000233

Diagram and typed description.

Question. There are compact



manif.

Describe regular differential operators

On the Riemann sphere $\frac{\partial}{\partial z},\ z\frac{\partial}{\partial z},\ z^2\frac{\partial}{\partial z}$

On Dixmier's work (Limanov).

$R_{1,0}$ — ring of quantum mech. — generators $x, \frac{d}{dx}$.

Diagram and typed description.



Polynomials

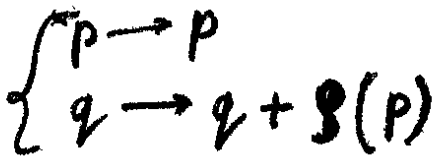
in p and q.

Original source details: GELFAND-P000233

$p, q, \quad [p, q] = 1.$

The autom. group is generated by triang. autom.

$$\begin{cases} p \to p \\ q \to q + f(p) \end{cases}$$



Uncertain reading. The letter before (p) is heavily inked. The reading f is provisional. The right-hand pair clearly has $g(q)$.

Original source details: GELFAND-P000233

and $\begin{cases} p \to p + g(q) \\ q \to q \end{cases}$

For the division ring $D_{1,0}$ unknown.

Or



Uncertain reading. A short word is crossed out. A possible reading is от, with the first letter obscured.

Original source details: GELFAND-P000233

$$\begin{cases} q \to q \\ p \to p + g(q) \end{cases}$$

and $\begin{cases} q \to p \\ p \to -q \end{cases}$

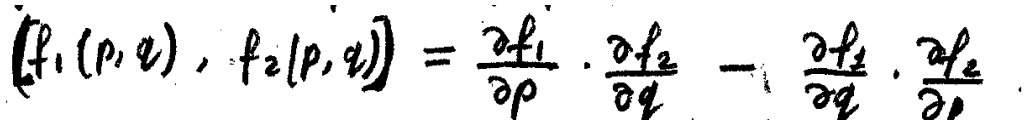
(Fourier transform)

Is generated

GELFAND-P000234

Arnold: theor.
Let there be functions $f(p,q)$

$$[f_1(p,q), f_2(p,q)] = \frac{\partial f_1}{\partial p}\cdot\frac{\partial f_2}{\partial q} - \frac{\partial f_2}{\partial q}\cdot\frac{\partial f_2}{\partial p}.$$



Original source details: GELFAND-P000234

Uncertain reading. The first numerator after the minus sign has a subscript with a foot, probably 2. The second numerator after the minus sign clearly has 2.

Consider 2 polynomials

$$P = P(p,q)$$
$$Q = Q(p,q).$$

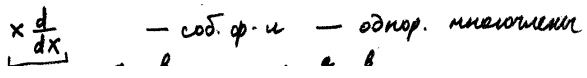
Substitute P and Q — will be 1.

They induce canon. transformations of quasiclassics.

But there is no connection, even with grading.

An element x is completely semisimple if the transformation ad X has a complete system of eigenvalues.

Then by an automorph. x is carried into $x \to pq$.



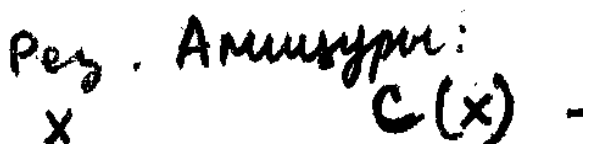
Original source details: GELFAND-P000234

Diagram and typed description.

$x\frac{d}{dx}$ — eigenfunctions — homog. polynomials.

$$p^a q^b \to (a-b)p^a q^b.$$

Result of Amitsur:



Original source details: GELFAND-P000234

Uncertain reading. The manuscript spelling probably ends -ури. An alternative is -ура.

x $C(x)$ — commuting — commut. subalgebra.
But these are not always polynomials.

$$\forall\alpha\ \exists X, Y, \qquad X^3 = Y^2 + \alpha, \qquad \alpha \in C.$$

Palamodov.

on Hörmander's work (on $\bar\partial$)

2 main results: existence theorem and approx. theorem.

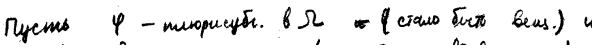
1. $\Omega \subset C^n$, $\phi \in C^2(\Omega)$

$$L(\phi) = \frac{\partial^2\phi}{\partial z_j \partial \bar z_k}\xi_j\bar\xi_k \geq 0$$

$$L_{p,q} = L_{p,q}(\Omega, \phi), \qquad \|f\|_\phi^2 = \int_\Omega |f|^2 e^{-\phi} dV$$

forms of type (p, q)

Let ϕ be plurisubh. in Ω



Original source details: GELFAND-P000234

Uncertain reading. A short dense stroke stands between Ω and the opening parenthesis. Its reading is unresolved.

(therefore real) and

$\forall c \{\phi \le c\}$ — compact. ($\Rightarrow \Omega$ — pseudoconvex)

Then $\forall f \in L_{p,q}$, $\bar\partial f = 0$

$\exists$

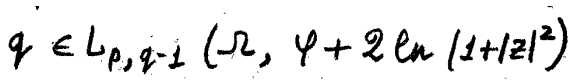


Original source details: GELFAND-P000234

Undeciphered passage. Crossed-out fragment; its initial symbol resembles g, the remaining letters have not been established.

g of type $p, q-1$

$$\bar\partial g = f \quad \overset{(q\ge 1)}{\text{and}} \quad g \in L_{p,q-1}(\Omega, \phi + 2\ln(1+|z|^2)$$



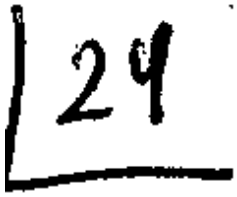
Original source details: GELFAND-P000234

Manuscript notation. One closing parenthesis is visible after $|z|^2$. The outer closing parenthesis is absent.

Here $\sqrt{q}\,\|g\| \le \|f\|$ (assuming $q \ge 1$)

GELFAND-P000235

[source number: 24]



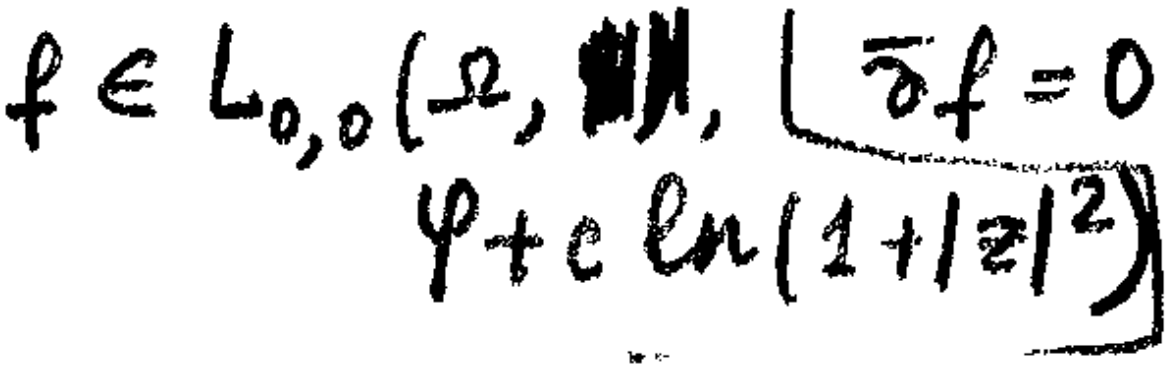
Original source details: GELFAND-P000235

Diagram and typed description.

24

2. Approx. theorem.

$\Omega = C^n$.



Original source details: GELFAND-P000235

Uncertain reading. The former second argument is crossed out: ϕ or ψ; $\phi + c\ln(1+|z|^2)$ is inserted below.

$f \in L_{0,0}(\Omega, [?])$ | $\bar\partial f = 0$ (holom. functions)

$\phi + c\ln(1+|z|^2)$

i.e. $|f| \le Ce^{\phi}(1+|z|^2)^k$, ($\exists C > 0$).

This set is nonempty (nontriv. — $\Omega = C^n$). — and dense in the space of all [holom.] entire functions with unif. convergence on $\forall$ compact set:

ϕ — homog. along radii $\Rightarrow$ indicator of an entire function.

Hence Martino: the class of indicators — all radially homog. plurisubh. harm.



Original source details: GELFAND-P000235

Uncertain reading.

$i(z) =$ | $\lim_{\tau\to\infty}$ | $\tau^{-k}\ln|f(z\tau)|$

A stroke above lim meets the left side of the box. Its function is uncertain.

18/III

If $\phi = \phi(x)$ (only on the real parts), it is plurisubharm. if and only if it is convex.

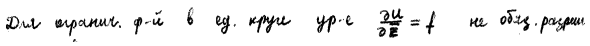
In $\forall$ one-dim. complex space subharm.

Example:

$$\frac{\partial u}{\partial \bar z} = f, \qquad |f(z)| \le Ce^{a|\operatorname{Im} z|}(1+|z|^2)^k.$$

A solution in this class — nontrivial.

For bounded functions in the unit disk, the equation $\dfrac{\partial u}{\partial z} = f$



Original source details: GELFAND-P000235

Uncertain reading. There are thick strokes between z and the fraction rule. No separate bar can be distinguished. Provisional z, possibly $\bar{z}$.

is not necessarily solvable for bounded f.

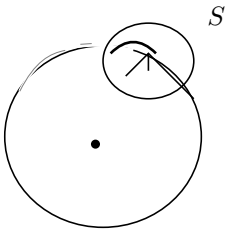
For solvability it suffices

(f — measure)



Original source details: GELFAND-P000235

Uncertain reading.



The faint upper arc and dark strokes at the intersection are transcribed without an established meaning for the strokes.

S — disk. $|f(S \cap \Omega)| \le C R_S$.

Theorem 3. More precise theorem for fixed q.

Let $\partial\Omega \in C^3$ and the domain be bounded. Local condition on the boundary:

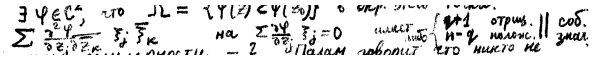
$\forall z \in \partial\Omega\ \exists \phi \in C^2$ such that $\Omega = \{\phi(z) < \phi(z_0)\}$ in a nbhd. of this point.

Condition a_q:

$$\sum \frac{\partial^2 \phi}{\partial z_j \partial \bar{z}_k} \xi_j \bar{\xi}_k \quad \text{on} \quad \sum \frac{\partial \phi}{\partial z_j} \xi_j = 0$$

has —

$$\begin{cases} q+1 \text{ negative} \\ \text{or } n-q \text{ positive} \end{cases} \quad \| \quad \text{eigen-values}$$



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Manuscript notation. The tangent condition uses ξ_j. At right the alternatives are $q+1$ negative or $n-q$ positive, followed by two strokes.

Gelfand: domains of regularity — ?

Palam. says nobody is working on this.

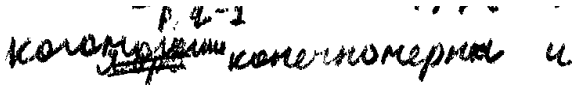
$\partial\Omega$ pseudoconvex if at $\forall$ point condition a_0 (at $\forall$ point $n-1$ positive eigenvalues)

In the tangent space, $n-q$-dim. surfaces along which the domain is pseudoconvex.

Let a_q hold at $\forall$ boundary point.

$$L_{p,q-1} \xrightarrow{\bar{\partial}} L_{p,q}(\Omega, \phi) \xrightarrow{\bar{\partial}} L_{p,q+1}$$

Cohomology



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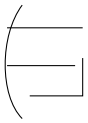
Uncertain reading. [crossed out below: kernel, ядро or ядра]

is finite-dimensional and the image is closed.



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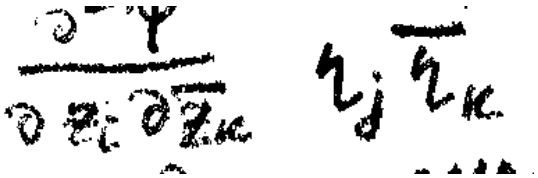
Uncertain reading.



Isolated sign: E or $\exists$; its function has not been established.

Existence of a function Ψ on the whole domain.
Ψ — real, defined inside Ω,

$$\frac{\partial^2 \Psi}{\partial z_i \partial \bar{z}_k} \eta_j \bar{\eta}_k$$



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Uncertain reading. The first denominator index is short with a dot, probably i. The η index has a descending loop and reads j.

satisfies all conditions

(1 more plus required, since in the whole space)

Then cohomology 0.

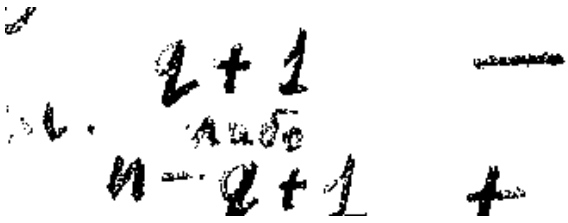
GELFAND-P000236

All these theorems for complex spaces — Andreotti and Grauert.

1. Runge-type theorem.

$\omega \subset \Omega$, $L(\Psi)$ at $\forall$ point of $\Omega \setminus \omega$ satisfies the condition

$$q+1 \quad -$$
$$\text{or}$$
$$n-q+1 \quad +$$



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Manuscript notation. The upper count is $q+1$ with a minus sign. The lower count is $n-q+1$ with a plus sign. The word between them is “or”.

$\Psi \to \infty$ near $\partial\Omega$, $\{\Psi < 0\} = \omega$

Then the cohomology of Ω restricts monomorphically to the cohom. of ω.

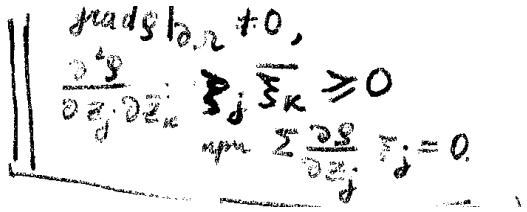
A closed form f in Ω and the equation $\bar{\partial}u = f$ solvable in $\omega \Rightarrow$ it is solvable in Ω as well.

2. Approximation theorem.

Closed forms in ω are approximated by closed forms in Ω.

Existence theorem.

Let $\partial\Omega \in C^2$ and there exist ρ, $\{\rho < 0\} = \Omega$,



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Diagram and typed description.

grad $\rho|_{\partial\Omega} \neq 0$,

$\frac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k} \xi_j \bar{\xi}_k \geq 0$

when $\sum \frac{\partial \rho}{\partial z_j} \xi_j = 0$.

It is proved simply, using the identity.

$$L_{p,q-1} \xrightarrow{\bar{\partial}=T} L_{p,q} \xrightarrow{\bar{\partial}=S} L_{p,q+1}$$

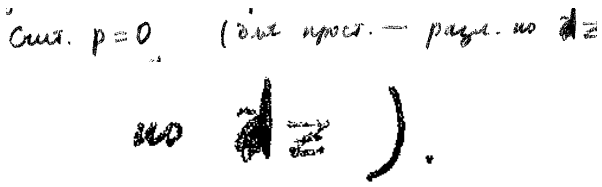
$$\|T^* f\|_\phi^2 + \|Sf\|_\phi^2 =$$

$$(u, v)_\phi = \int_\Omega u \bar{v} e^{-\phi} dV.$$

(adjoint with respect to this inner product)

$$= \sum_L \sum_{j,k} \left(\frac{\partial^2 \phi}{\partial z_j \partial \bar{z}_k} f_{jL}, f_{kL} \right)_\phi + \sum_J \sum_k \left\| \frac{\partial f_J}{\partial \bar{z}_k} \right\|_\phi^2 +$$

Take $p = 0$ (for simplicity — expansion in dz?).



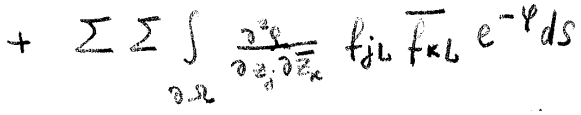
Uncertain reading. The first character is overwritten and probably reads d. A separate bar over z is uncertain. The next formula clearly has $d\bar{z}^J$.

Original source details: GELFAND-P000236

$$f = \sum f_J d\bar{z}^J, \qquad J = j_1, \ldots, j_q.$$

f_J skew-symm. in the indices, jL — long index.

$$+ \sum \sum \int_{\partial\Omega} \frac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k} f_{jL} \overline{f_{kL}} e^{-\phi} dS.$$



Manuscript notation. Two summation signs stand before the integral without lower indices. The integral is over $\partial\Omega$ and ends in dS.

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The identity — integration by parts.

First — Morrey with $\phi = 0$.

grad, rot — known in mechanics.

Solvability follows from it.

Next time — Nirenberg (according to hearsay).